深度学习平台与应用

第十一讲：视频理解

计算机视觉任务

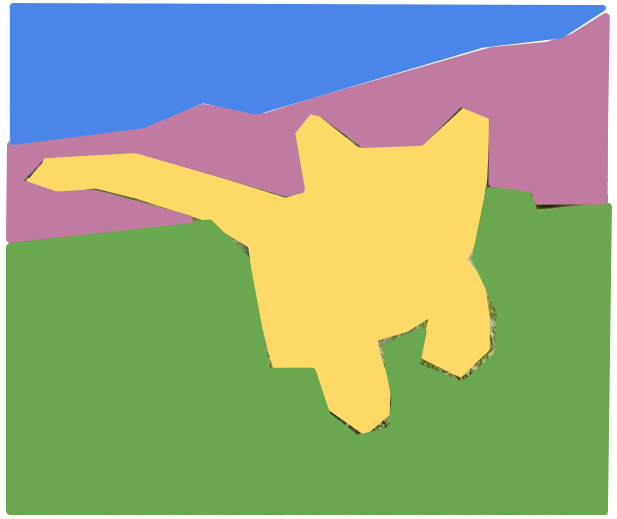
- **分类、分割、检测**

Classification

CAT

No spatial extent

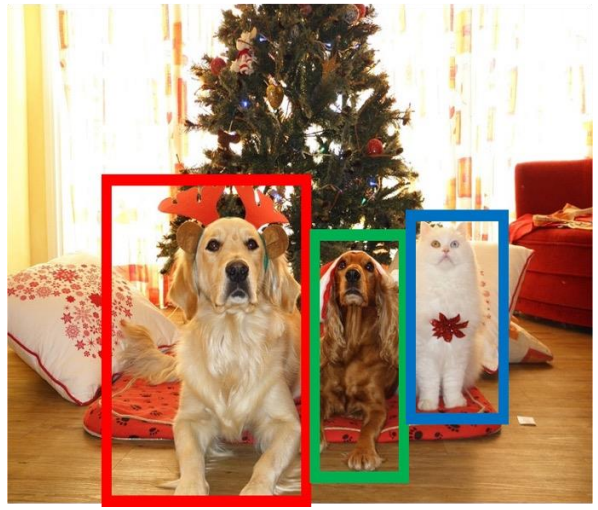
Semantic Segmentation



GRASS, CAT, TREE, SKY

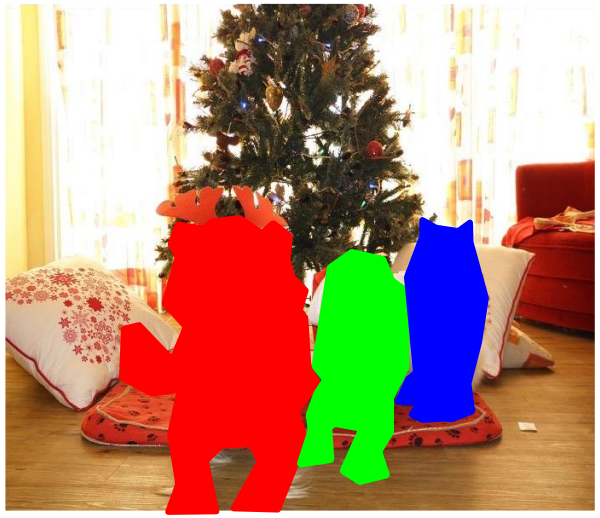
No objects, just pixels

Object Detection



DOG, DOG, CAT

Instance Segmentation

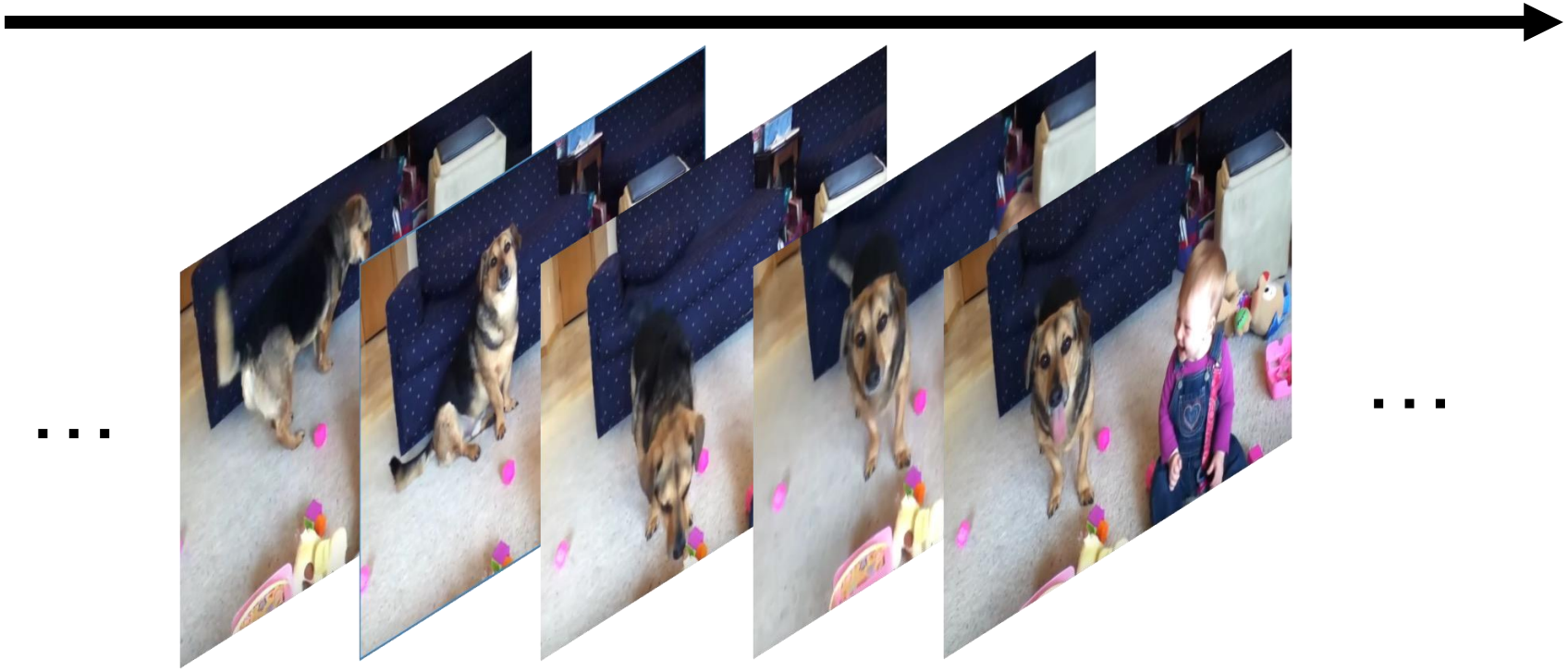


DOG, DOG, CAT

Multiple Object

视频理解

- Video = 2D + Time
- 视频由图片序列组成：T x 3 x H x W



视频理解

- **视频分类**

Input video:
T x 3 x H x W

Swimming
Running
Jumping
Eating
Standing

视频理解

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视频分类

图片：识别类别

Dog
Cat
Fish
Truck

视频：识别动作

Swimming
Running
Jumping
Eating
Standing

视频理解

- 问题：视频太大了！
- 30 FPS 的视频：
  - SD (640 x 480): 每分钟 ~1.5 GB
  - HD (1920 x 1080): 每分钟 ~10 GB

Input video:
T x 3 x H x W

视频理解

- 问题：视频太大了！
- 30 FPS 的视频：
  - SD (640 x 480): 每分钟 ~1.5 GB
  - HD (1920 x 1080): 每分钟 ~10 GB
- 解决方案：
  - 在切片（clips）上训练，低 FPS，低分辨率
  - T=16，H=W=112 时，3.2s 视频 588KB 大小

Input video:
T x 3 x H x W

视频理解

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- **视频切片（clips）训练**

原始视频：长视频, 高 FPS

视频理解

南京大学 NANJING UNIVERSITY

- **视频切片（clips）训练**

原始视频：长视频, 高 FPS

训练：在低 FPS 的 clips 上训练

视频理解

南京大学 NANJING UNIVERSITY

- **视频切片（clips）训练**

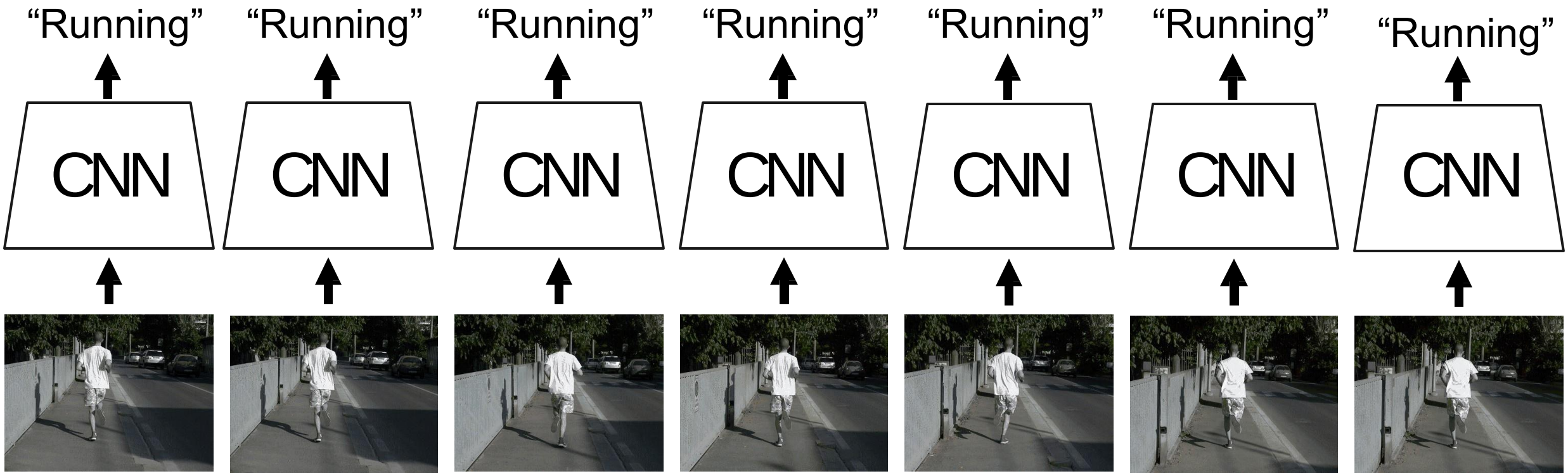
原始视频：长视频，高 FPS

训练：在低 FPS 的 clips 上训练

测试：在不同 clips 上运行模型，取平均预测结果

视频理解

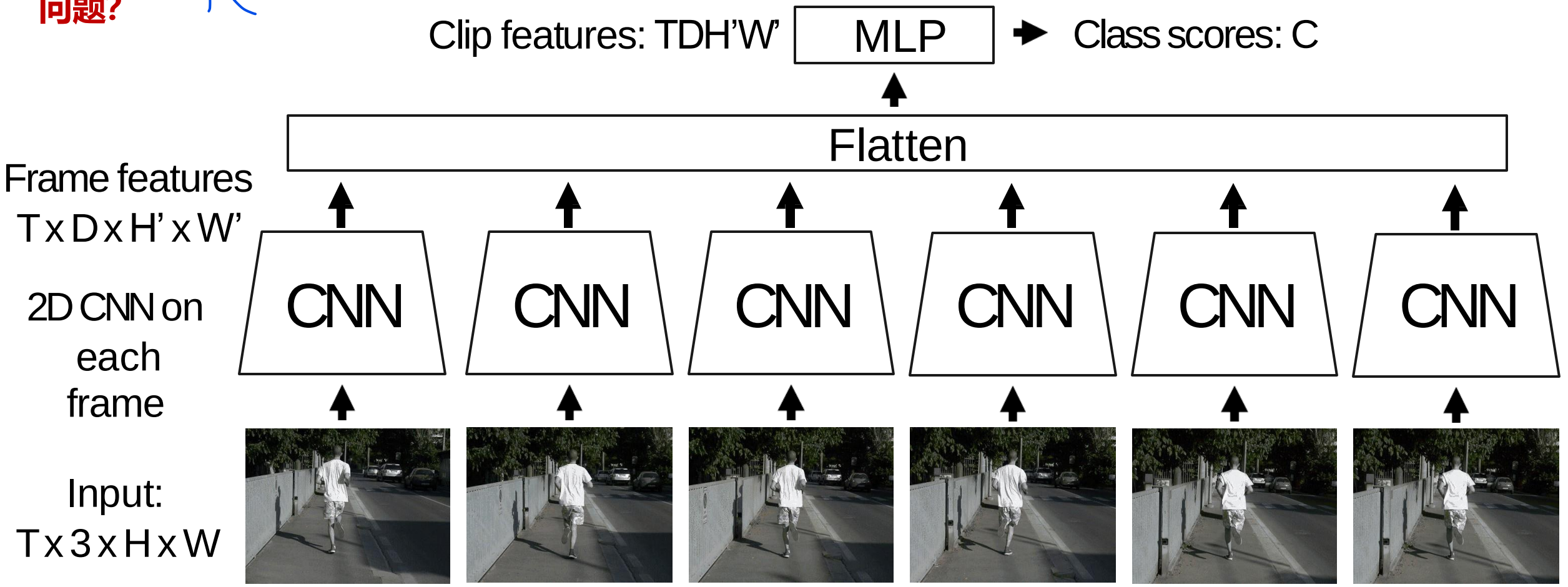
- 单帧 CNN 模型：训练普通的 2D CNN 独立对视频帧进行分类
- 通常是视频分类的一个非常强的基线方法



视频理解

- **Late Fusion（FC）：将每帧的 CNN 特征并联并送入 MLP**

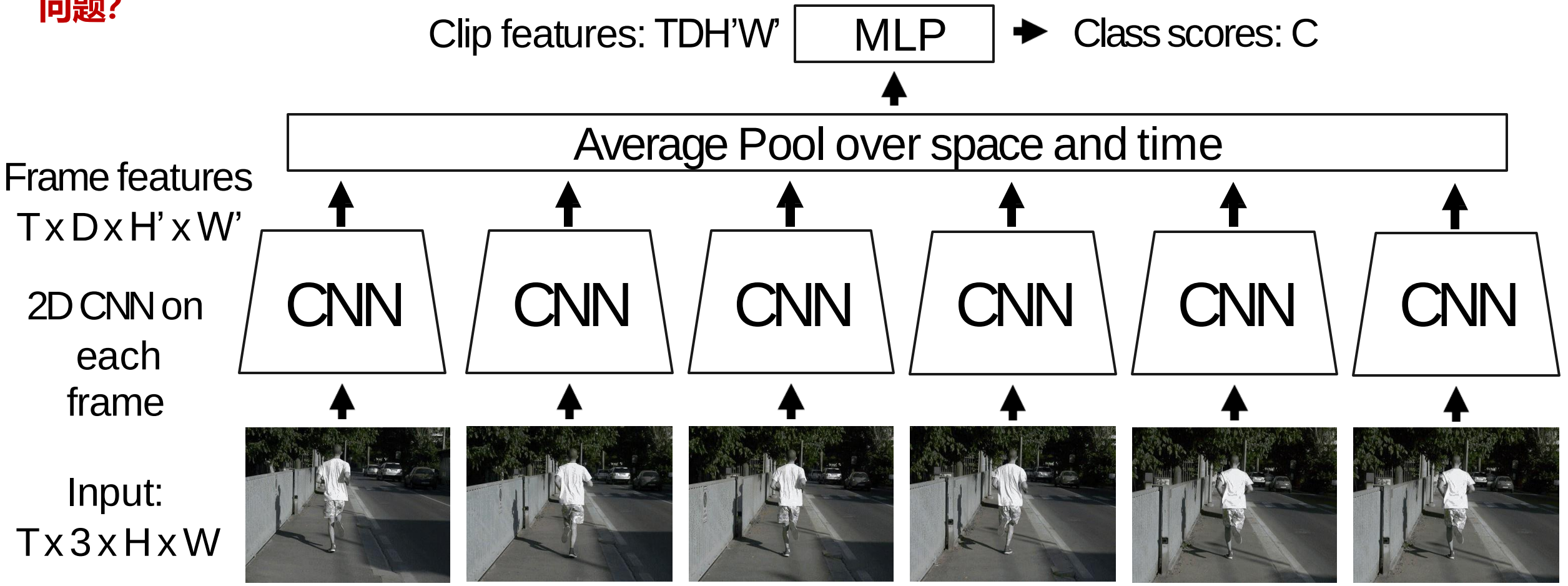
问题?



视频理解

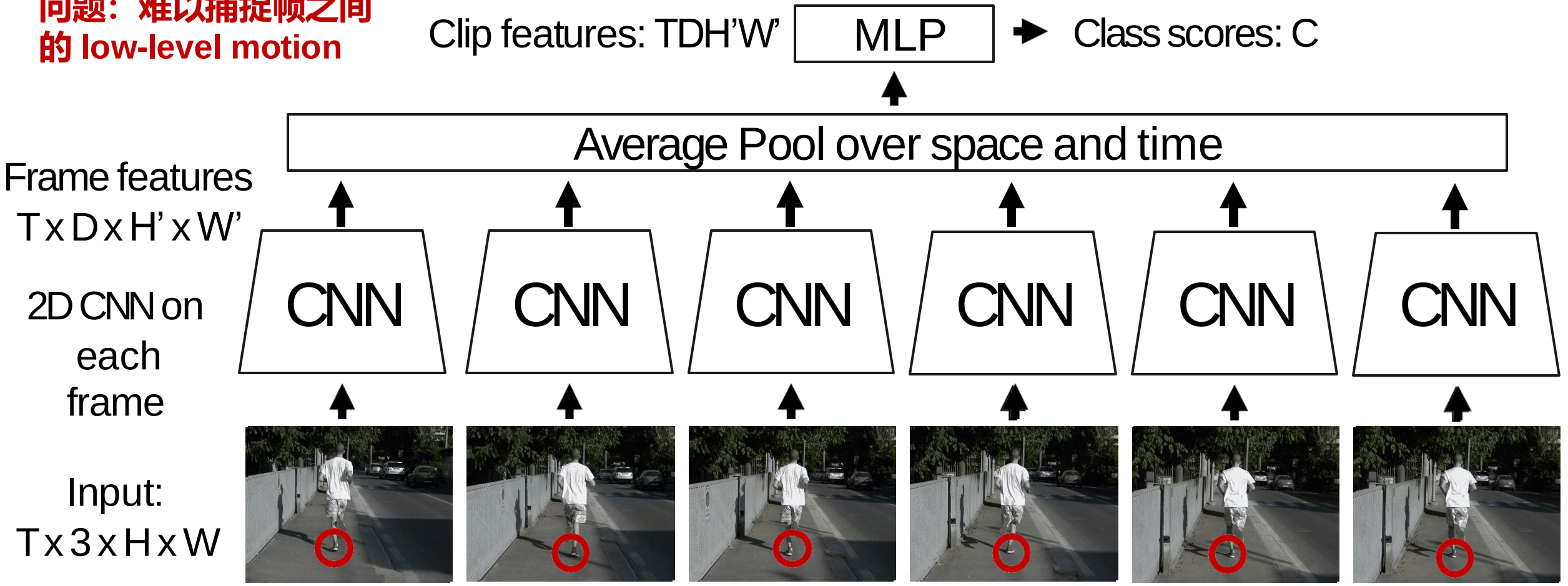
- **Late Fusion（Pooling）：将每帧 CNN 特征池化并送入线性层**

问题?



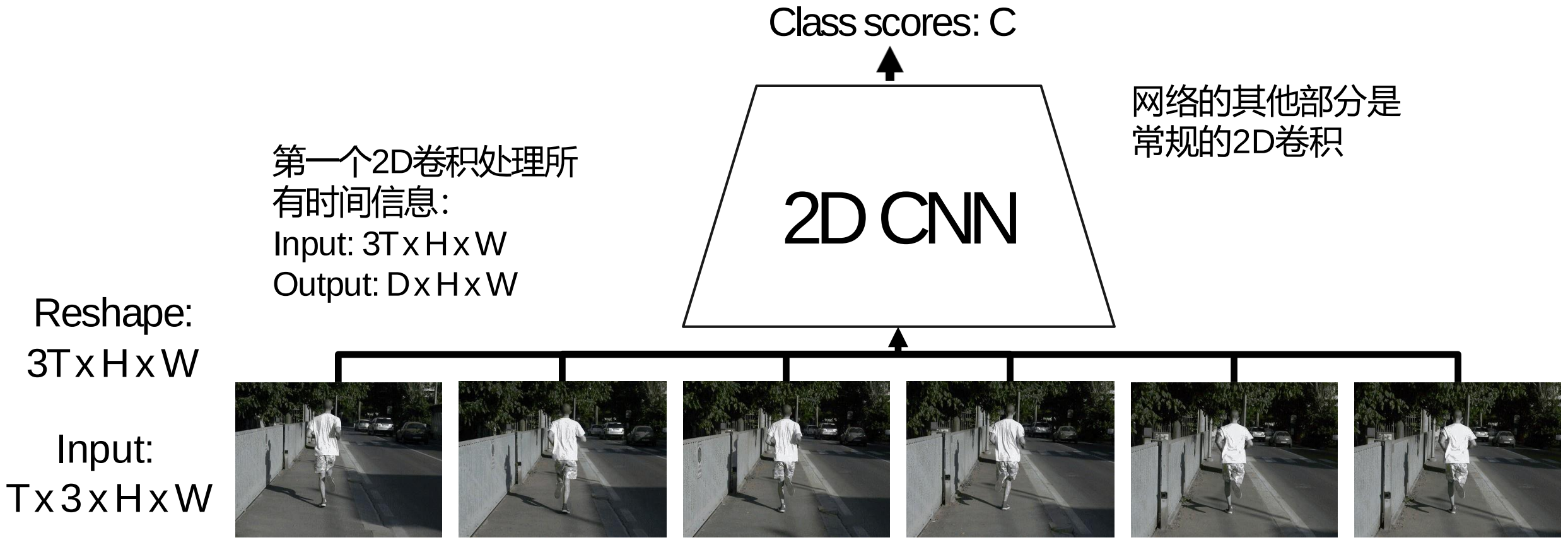
- **Late Fusion（Pooling）：将每帧 CNN 特征池化并送入线性层**

问题：难以捕捉帧之间的 low-level motion



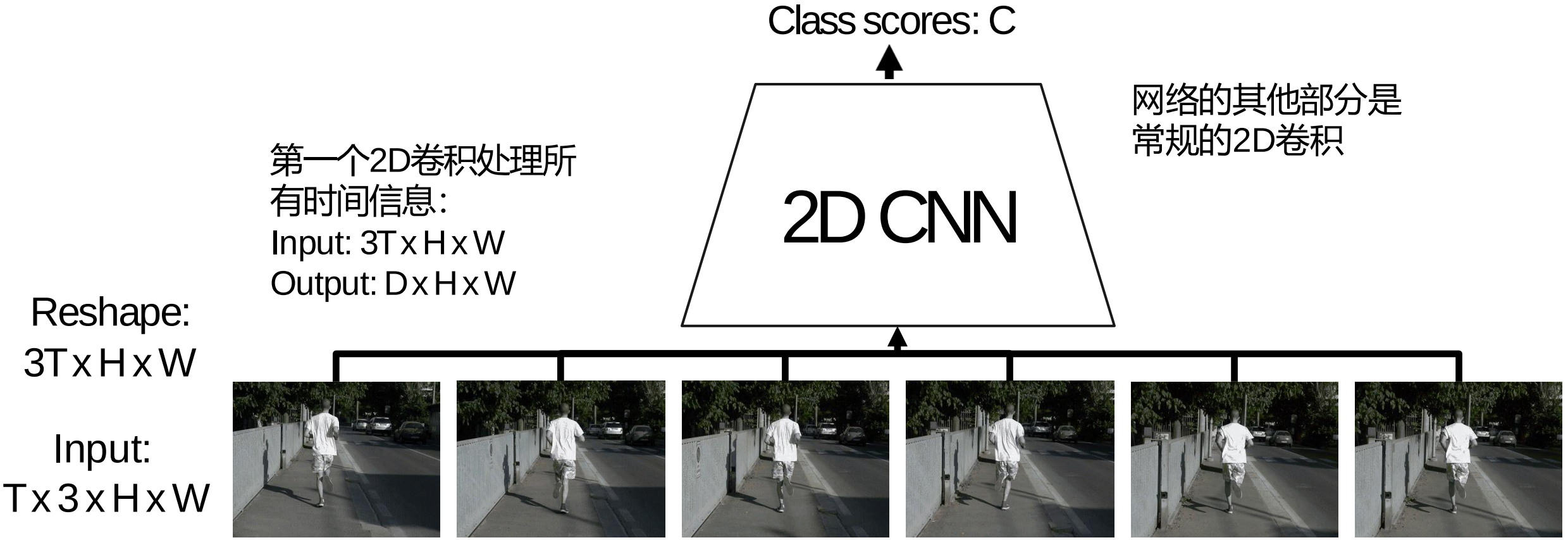
Early Fusion

问题?

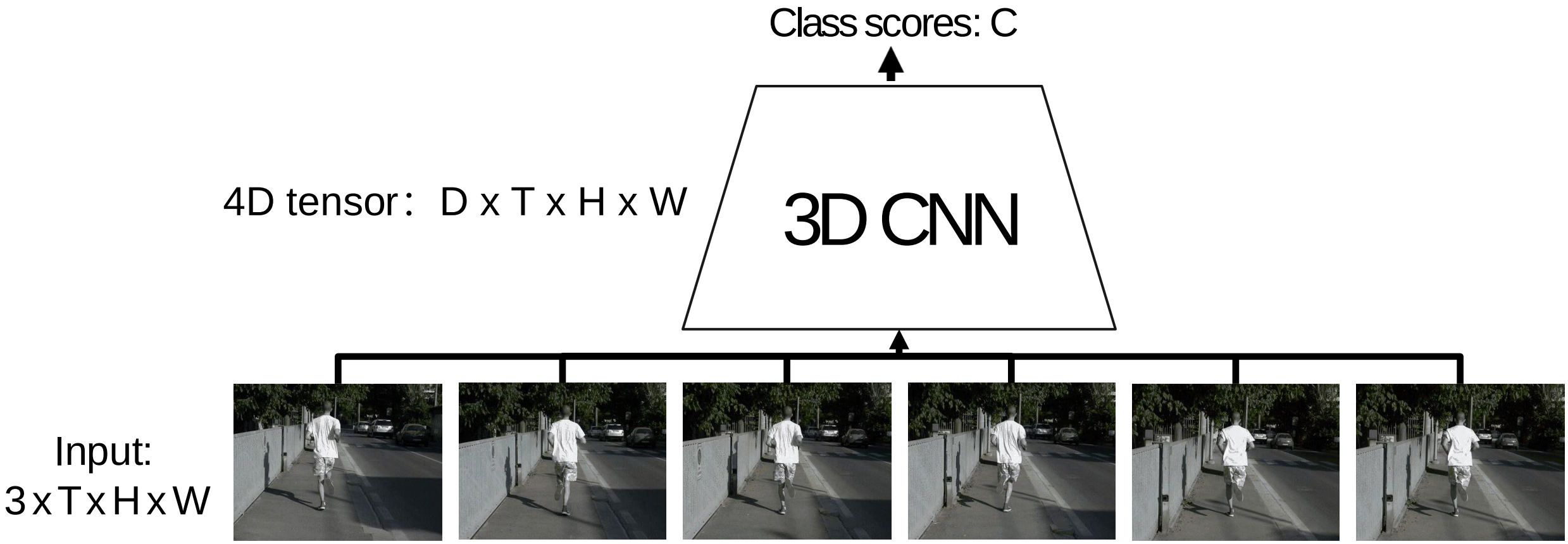


视频理解

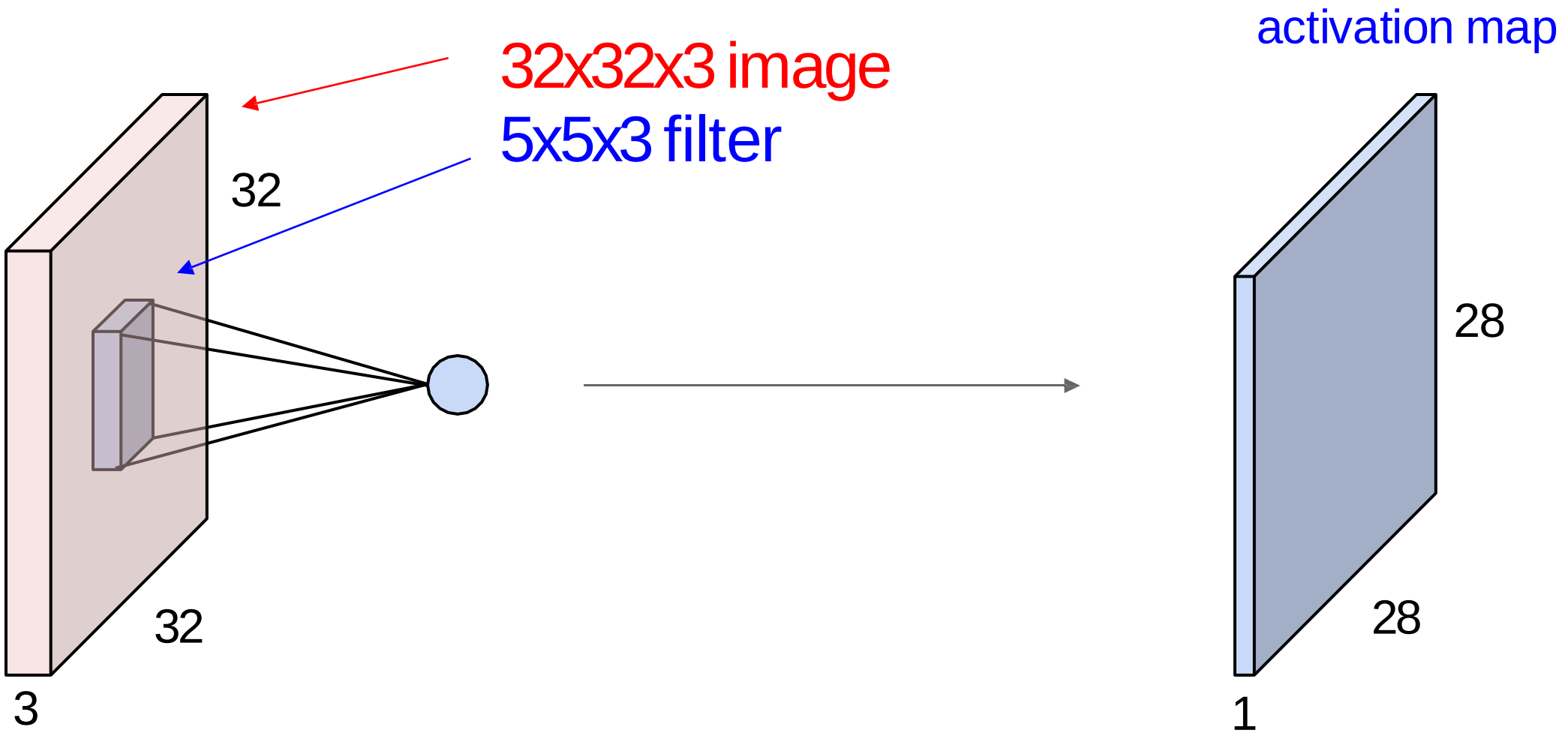
- **Early Fusion**
- **问题：一层卷积处理输入视频可能不够**



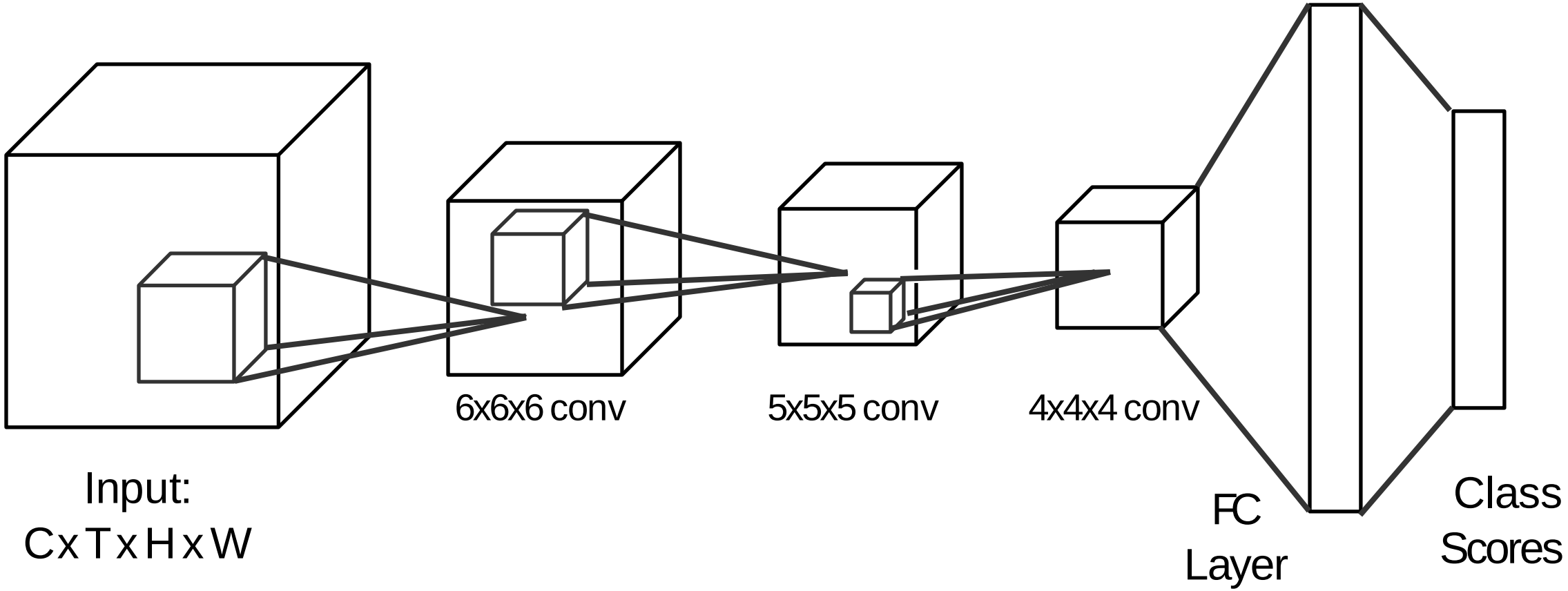
3D 卷积



2D 卷积



3D 卷积


6x6x6 conv
5x5x5 conv
4x4x4 conv
Input:
C x T x H x W
FC
Layer
Class
Scores

Early Fusion v.s. Late Fusion v.s. 3D CNN

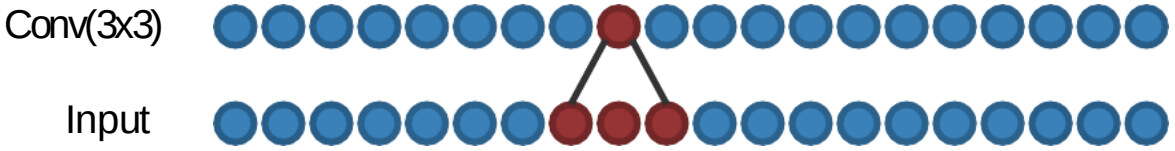
Late Fusion

Layer	Size **(C x T x H x W)**	Receptive Field **(T x H x W)**
Input	3 x 20 x 64 x 64	
Conv2D(3x3, 3->12)	12 x 20 x 64 x 64	1 x 3 x 3

Early Fusion v.s. Late Fusion v.s. 3D CNN

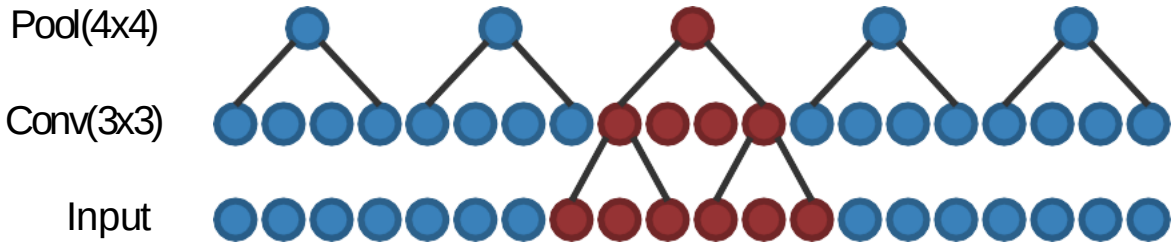
Late Fusion

Layer	Size **(C x T x H x W)**	Receptive Field **(T x H x W)**
Input	3 x 20 x 64 x 64	
Conv2D(3x3, 3->12)	12 x 20 x 64 x 64	1 x 3 x 3



Early Fusion v.s. Late Fusion v.s. 3D CNN

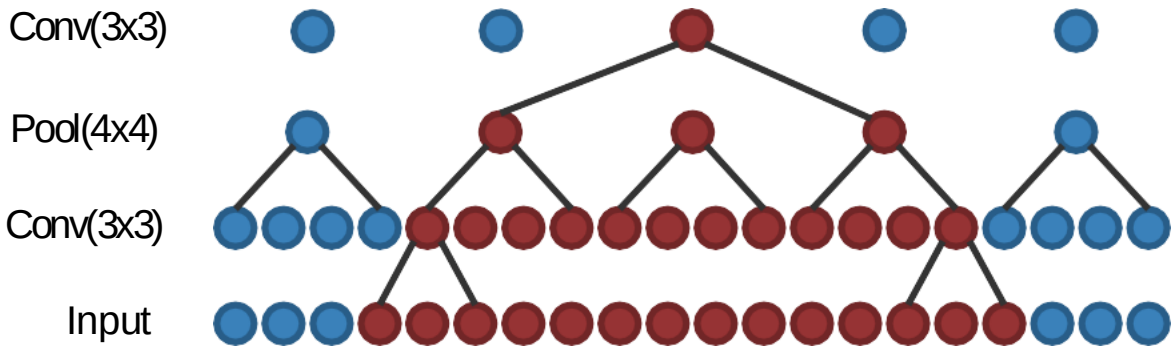
	Layer	Size **(C x T x H x W)**	Receptive Field **(T x H x W)**
Late Fusion	Input	3 x 20 x 64 x 64	
	Conv2D(3x3, 3->12)	12 x 20 x 64 x 64	1 x 3 x 3
	Pool2D(4x4)	12 x 20 x 16 x 16	1 x 6 x 6



Early Fusion v.s. Late Fusion v.s. 3D CNN

Late Fusion

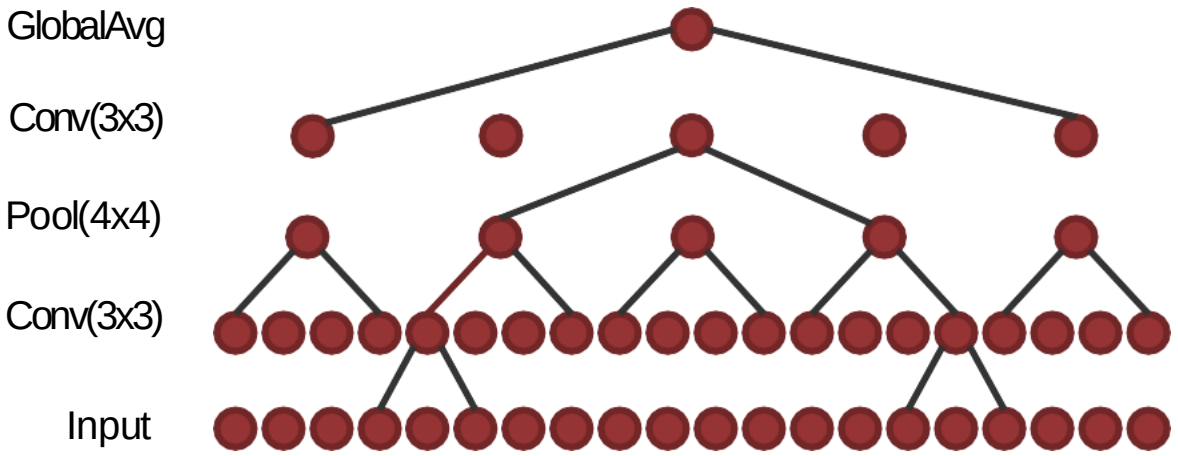
Layer	Size **(C x T x H x W)**	Receptive Field **(T x H x W)**
Input	3 x 20 x 64 x 64	
Conv2D(3x3, 3->12)	12 x 20 x 64 x 64	1 x 3 x 3
Pool2D(4x4)	12 x 20 x 16 x 16	1 x 6 x 6
Conv2D(3x3, 12->24)	24 x 20 x 16 x 16	1 x 14 x 14



Early Fusion v.s. Late Fusion v.s. 3D CNN

Late Fusion

Layer	Size **(C x T x H x W)**	Receptive Field **(T x H x W)**
Input	3 x 20 x 64 x 64	
Conv2D(3x3, 3->12)	12 x 20 x 64 x 64	1 x 3 x 3
Pool2D(4x4)	12 x 20 x 16 x 16	1 x 6 x 6
Conv2D(3x3, 12->24)	24 x 20 x 16 x 16	1 x 14 x 14
GlobalAvgPool	24 x 1 x 1 x 1	20 x 64 x 64



视频理解

	Layer	Size **(C x T x H x W)**	Receptive Field **(T x H x W)**
Late Fusion	Input	3 x 20 x 64 x 64	
	Conv2D(3x3, 3->12)	12 x 20 x 64 x 64	1 x 3 x 3
	Pool2D(4x4)	12 x 20 x 16 x 16	1 x 6 x 6
	Conv2D(3x3, 12->24)	24 x 20 x 16 x 16	1 x 14 x 14
	GlobalAvgPool	24 x 1 x 1 x 1	20 x 64 x 64
Early Fusion	Input	3 x 20 x 64 x 64	
	Conv2D(3x3, 3*20->12)	12 x 64 x 64	20 x 3 x 3
	Pool2D(4x4)	12 x 16 x 16	20 x 6 x 6
	Conv2D(3x3, 12->24)	24 x 16 x 16	20 x 14 x 14
	GlobalAvgPool	24 x 1 x 1	20 x 64 x 64
3D CNN	Input	3 x 20 x 64 x 64	
	Conv3D(3x3x3, 3->12)	12 x 20 x 64 x 64	3 x 3 x 3
	Pool3D(4x4x4)	12 x 5 x 16 x 16	6 x 6 x 6
	Conv3D(3x3x3, 12->24)	24 x 5 x 16 x 16	14 x 14 x 14
	GlobalAvgPool	24 x 1 x 1	20 x 64 x 64

视频理解

	Layer	Size (C x T x H x W)	Receptive Field (T x H x W)
Late Fusion	Input	3 x 20 x 64 x 64	
	Conv2D(3x3, 3->12)	12 x 20 x 64 x 64	1 x 3 x 3
	Pool2D(4x4)	12 x 20 x 16 x 16	1 x 6 x 6
	Conv2D(3x3, 12->24)	24 x 20 x 16 x 16	1 x 14 x 14
	GlobalAvgPool	24 x 1 x 1 x 1	20 x 64 x 64
Early Fusion	Input	3 x 20 x 64 x 64	
	Conv2D(3x3, 3*20->12)	12 x 64 x 64	20 x 3 x 3
	Pool2D(4x4)	12 x 16 x 16	20 x 6 x 6
	Conv2D(3x3, 12->24)	24 x 16 x 16	20 x 14 x 14
	GlobalAvgPool	24 x 1 x 1	20 x 64 x 64
3D CNN	Input	3 x 20 x 64 x 64	
	Conv3D(3x3x3, 3->12)	12 x 20 x 64 x 64	3 x 3 x 3
	Pool3D(4x4x4)	12 x 5 x 16 x 16	6 x 6 x 6
	Conv3D(3x3x3, 12->24)	24 x 5 x 16 x 16	14 x 14 x 14
	GlobalAvgPool	24 x 1 x 1	20 x 64 x 64

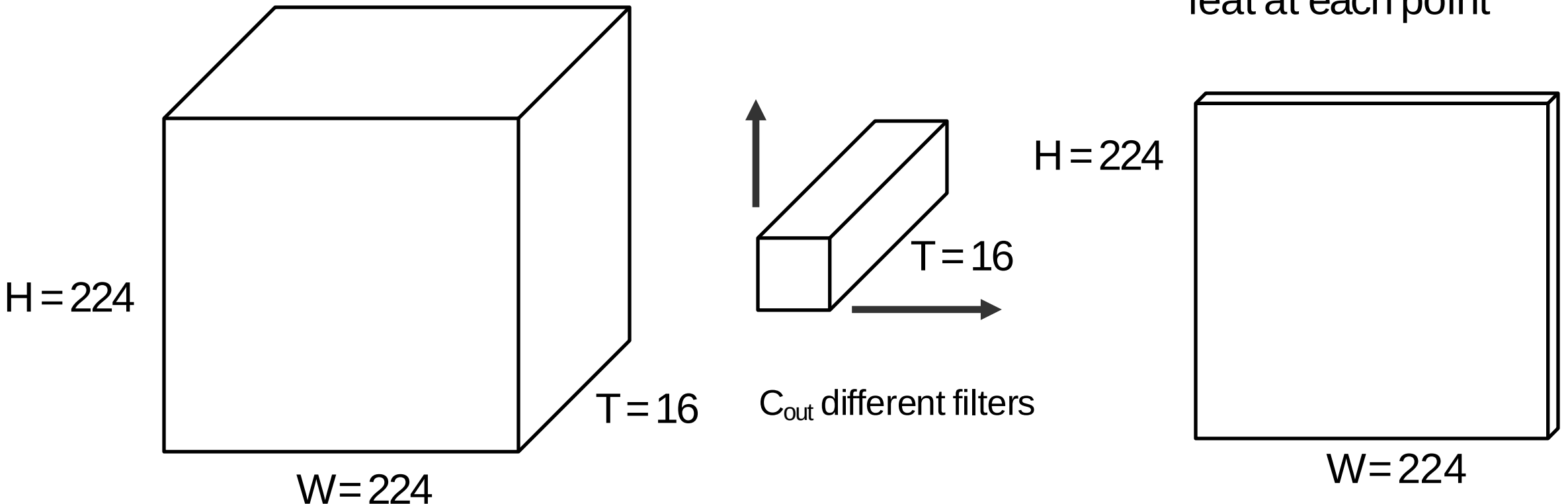
它们的区别是什么?

2D Conv (Early Fusion)

Input: C_{in} x T x H x W
(3D grid with C_{in}-dim feat at each point)

Weight:
C_{out} x C_{in} x (T x 3 x 3)
Slide over x and y

Output:
C_{out} x H x W
2D grid with C_{out} – dim feat at each point

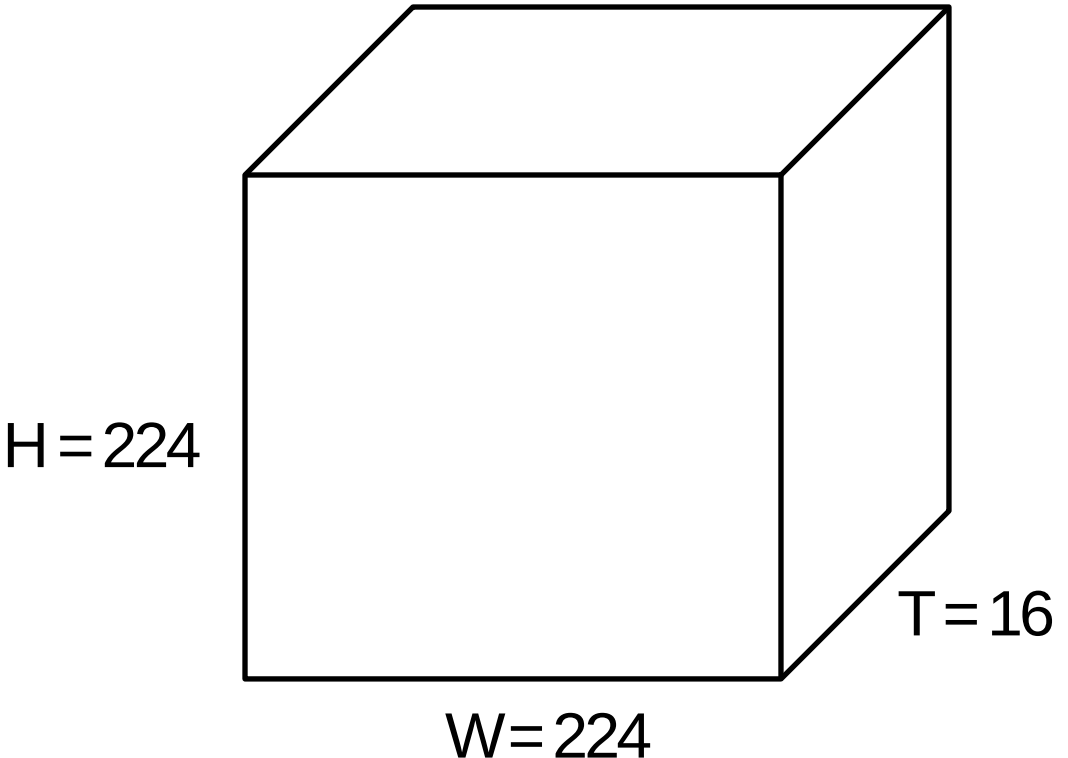


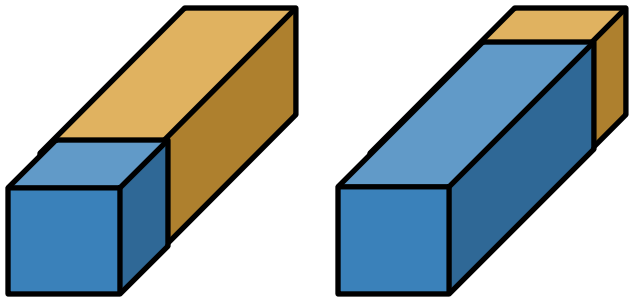
2D Conv (Early Fusion)

Input: C_{in} x T x H x W
(3D grid with C_{in}-dim feat at each point)

Weight:
C_{out} x C_{in} x (T x 3 x 3)
Slide over x and y

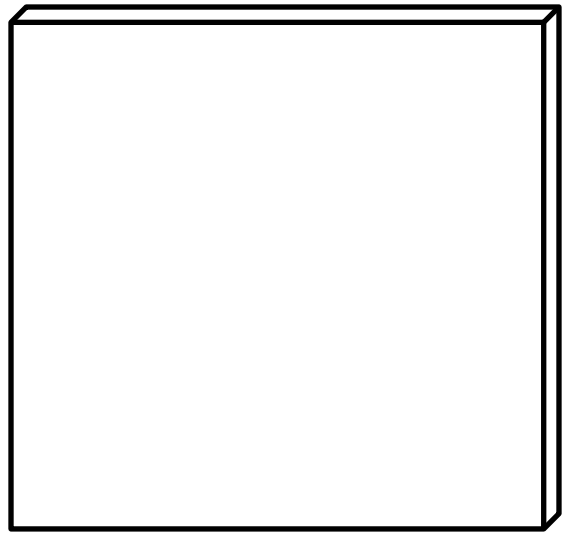
Output:
C_{out} x H x W
2D grid with C_{out} – dim feat at each point





C_{out} different filters

没有时间平移不变性！
不同时间都有单独滤波器器



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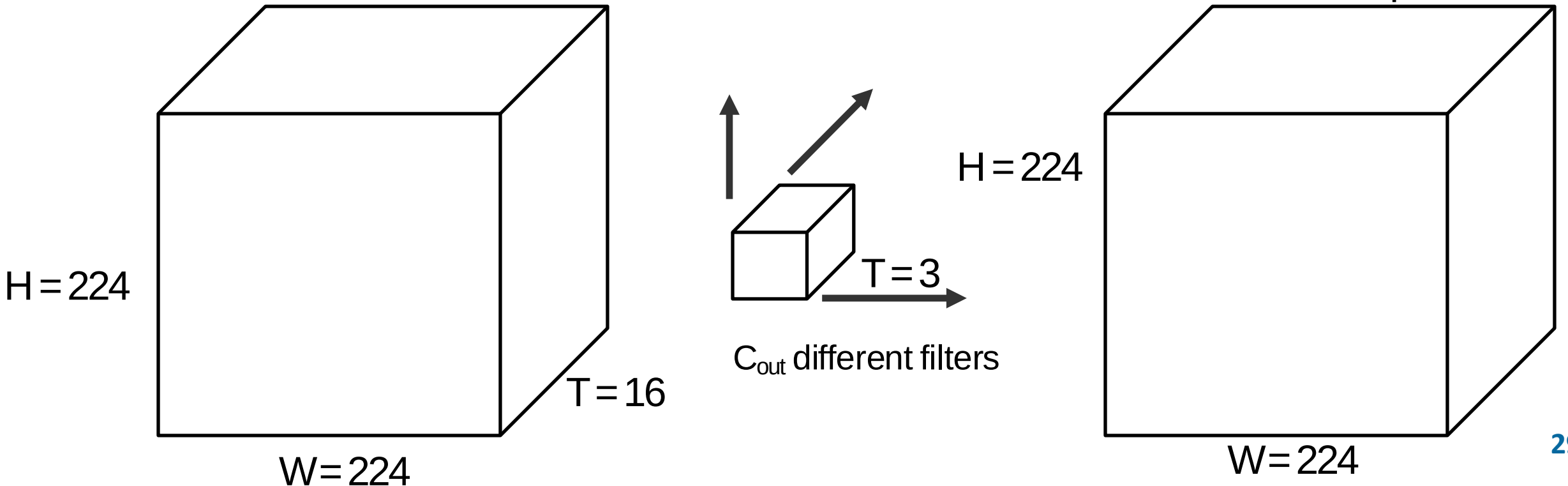
3D Conv

Input: C_{in} x T x H x W
(3D grid with C_{in}-dim feat at each point)

Weight:
C_{out} x C_{in} x (3 x 3 x 3)
Slide over x and y

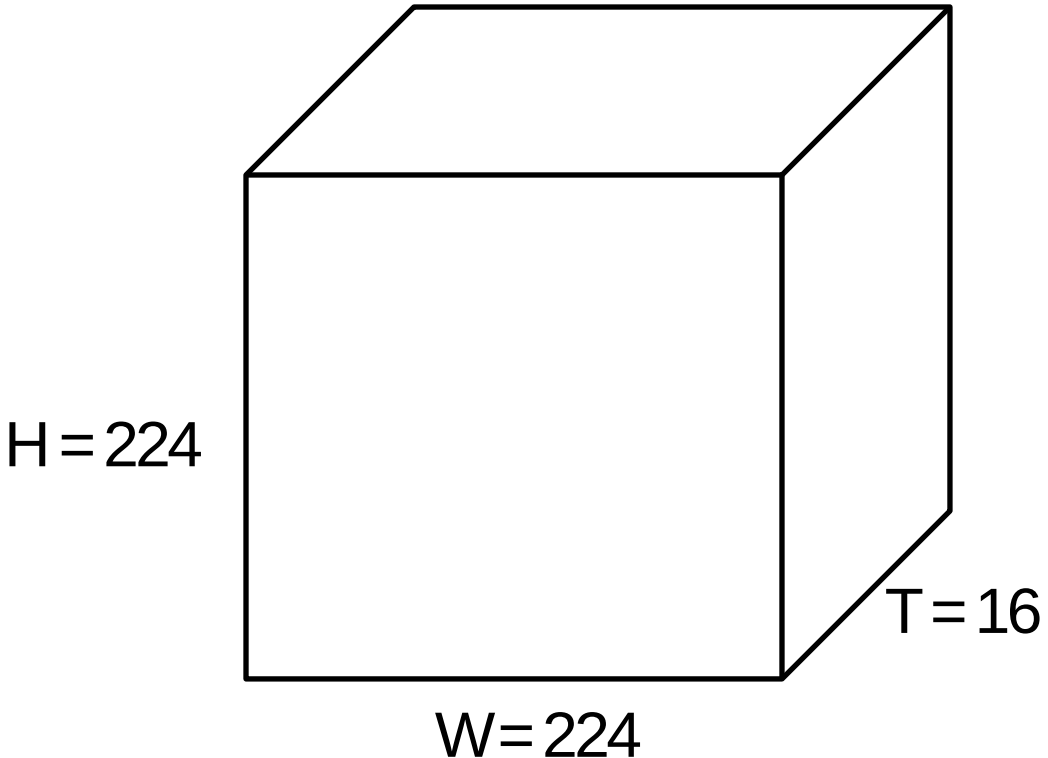
Output:
C_{out} x T x H x W
3D grid with C_{out}–dim feat at each point



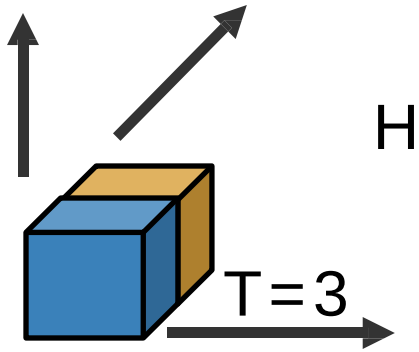
C_{out} different filters

3D Conv

Input: C_{in} x T x H x W
(3D grid with C_{in}-dim
feat at each point)



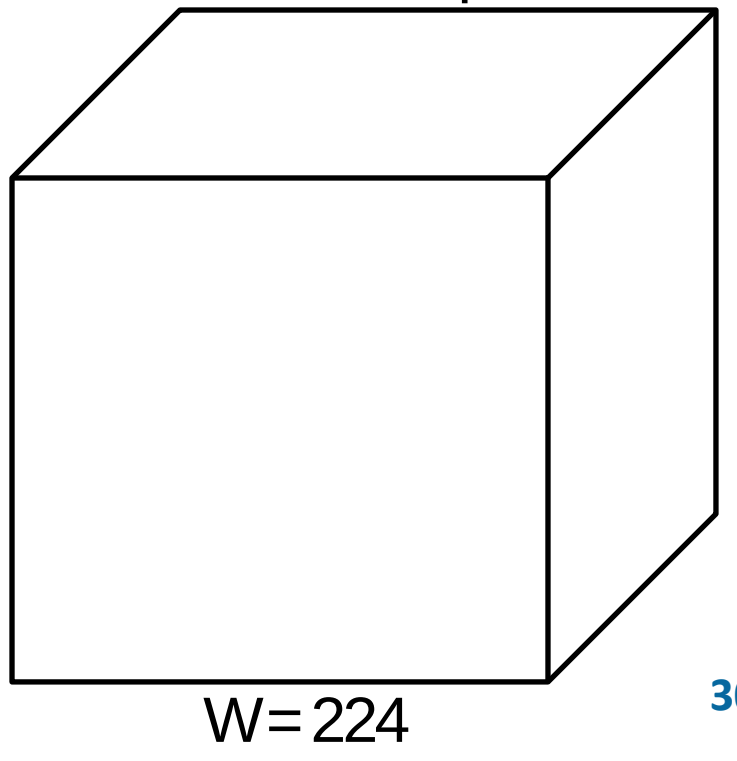
Weight:
C_{out} x C_{in} x (3 x 3 x 3)
Slide over x and y



C_{out} different filters

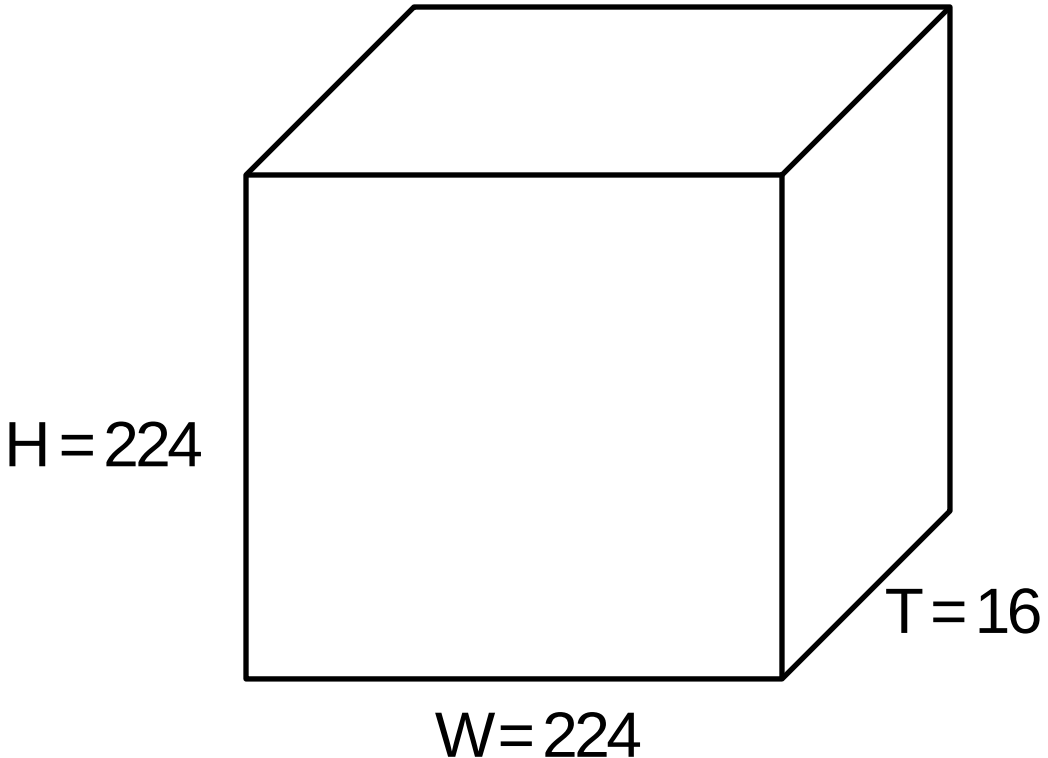
具有时间平移不变性！

Output:
C_{out} x T x H x W
3D grid with C_{out}–dim
feat at each point

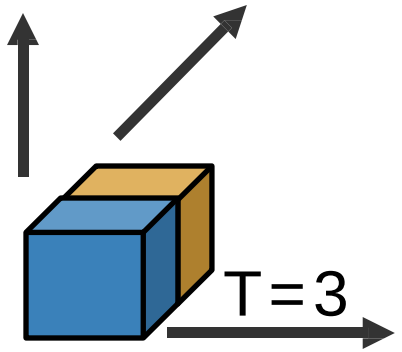


3D Conv

Input: C_{in} x T x H x W
(3D grid with C_{in}-dim feat at each point)



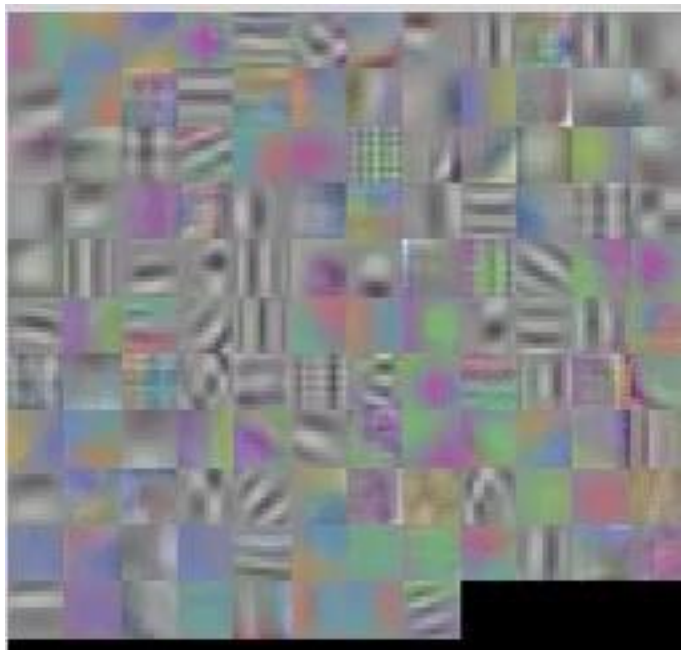
Weight:
C_{out} x C_{in} x (3 x 3 x 3)
Slide over x and y



C_{out} different filters

具有时间平移不变性！

第一层滤波器的形状为
3（RGB）x 4（帧）x 5 x 5（空间）



视频理解

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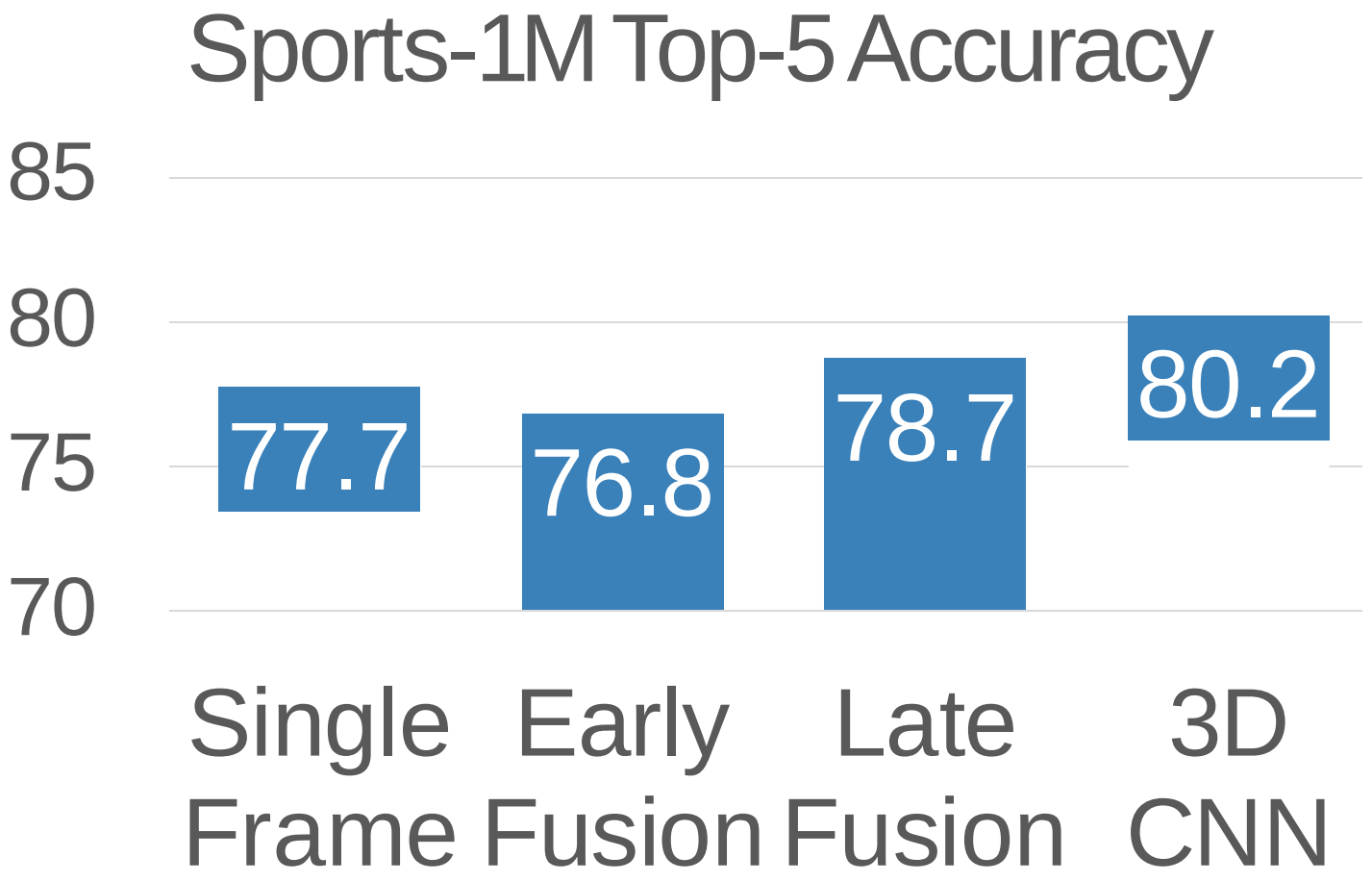
- 视频数据集：Sports-1M

100万个带有487种不同运动标签的YouTube视频

Ground Truth
Correct prediction
Incorrect prediction

- **Early Fusion v.s. Late Fusion v.s. 3D CNN**



单帧模型效果很好

3D模型提升很多

视频理解

- C3D：3D CNN 中的 VGG
  - 只使用 3x3x3 conv 和池化
  - 在Sports-1M上预训练

Layer	Size
Input	3 x 16 x 112 x 112
Conv1 (3x3x3)	64 x 16 x 112 x 112
Pool1 (1x2x2)	64 x 16 x 56 x 56
Conv2 (3x3x3)	128 x 16 x 56 x 56
Pool2 (2x2x2)	128 x 8 x 28 x 28
Conv3a (3x3x3)	256 x 8 x 28 x 28
Conv3b (3x3x3)	256 x 8 x 28 x 28
Pool3 (2x2x2)	256 x 4 x 14 x 14
Conv4a (3x3x3)	512 x 4 x 14 x 14
Conv4b (3x3x3)	512 x 4 x 14 x 14
Pool4 (2x2x2)	512 x 2 x 7 x 7
Conv5a (3x3x3)	512 x 2 x 7 x 7
Conv5b (3x3x3)	512 x 2 x 7 x 7
Pool5	512 x 1 x 3 x 3
FC6	4096
FC7	4096
FC8	C

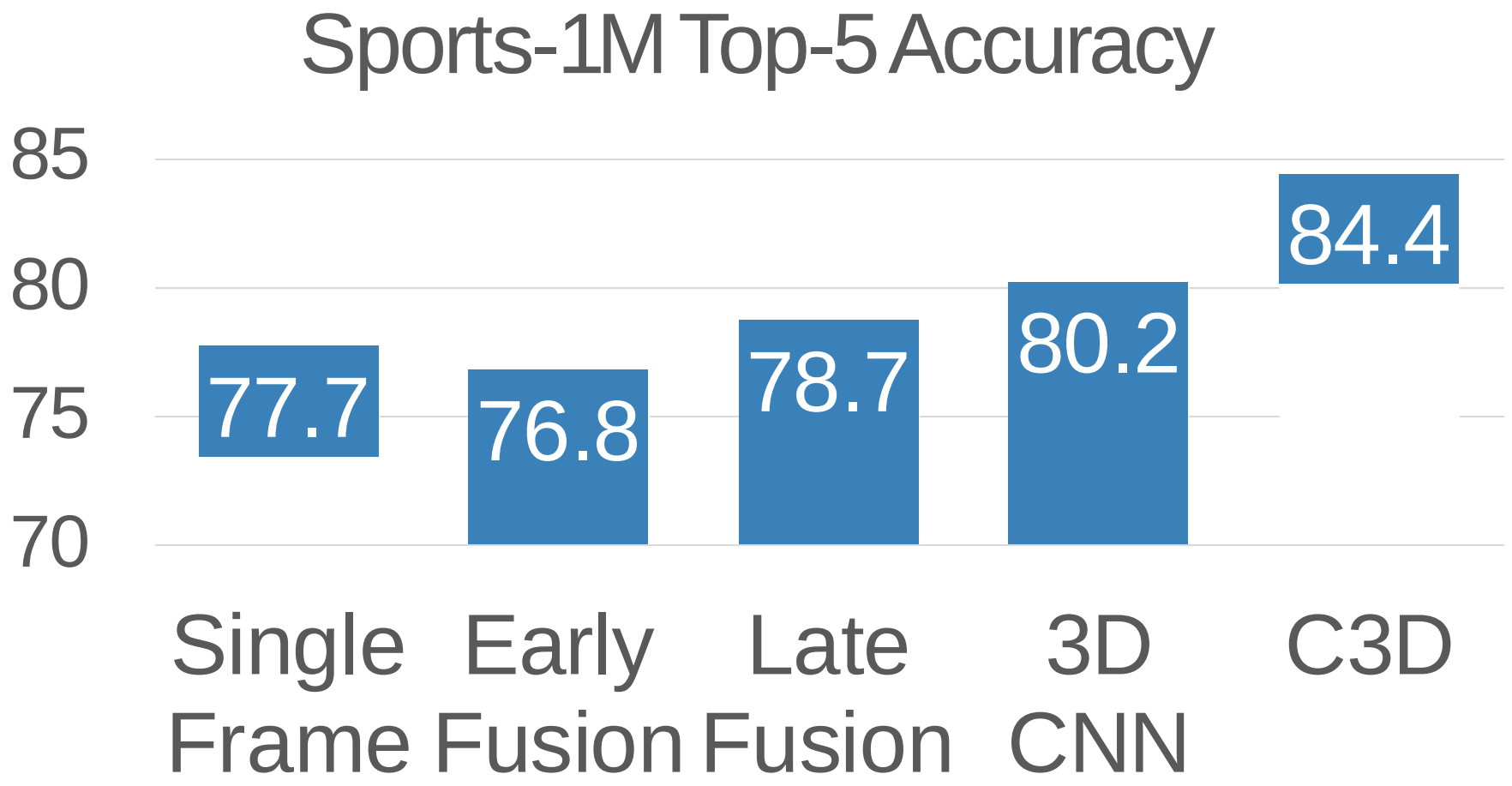
视频理解

- C3D：3D CNN 中的 VGG
  - 只使用 3x3x3 conv 和池化
  - 在Sports-1M上预训练
- 问题：3x3x3 conv 计算量很大
  - AlexNet: 0.7 GFLOP
  - VGG-16: 13.6 GFLOP
  - C3D: 39.5 GFLOP (2.9x VGG!)

Layer	Size	MFLOPs
Input	3 x 16 x 112 x 112	
Conv1 (3x3x3)	64 x 16 x 112 x 112	1.04
Pool1 (1x2x2)	64 x 16 x 56 x 56	
Conv2 (3x3x3)	128 x 16 x 56 x 56	11.10
Pool2 (2x2x2)	128 x 8 x 28 x 28	
Conv3a (3x3x3)	256 x 8 x 28 x 28	5.55
Conv3b (3x3x3)	256 x 8 x 28 x 28	11.10
Pool3 (2x2x2)	256 x 4 x 14 x 14	
Conv4a (3x3x3)	512 x 4 x 14 x 14	2.77
Conv4b (3x3x3)	512 x 4 x 14 x 14	5.55
Pool4 (2x2x2)	512 x 2 x 7 x 7	
Conv5a (3x3x3)	512 x 2 x 7 x 7	0.69
Conv5b (3x3x3)	512 x 2 x 7 x 7	0.69
Pool5	512 x 1 x 3 x 3	
FC6	4096	0.51
FC7	4096	0.45
FC8	C	0.05

视频理解

- **Early Fusion v.s. Late Fusion v.s. 3D CNN**



衡量运动（Motion）：光流（Optical Flow）

Image at frame t

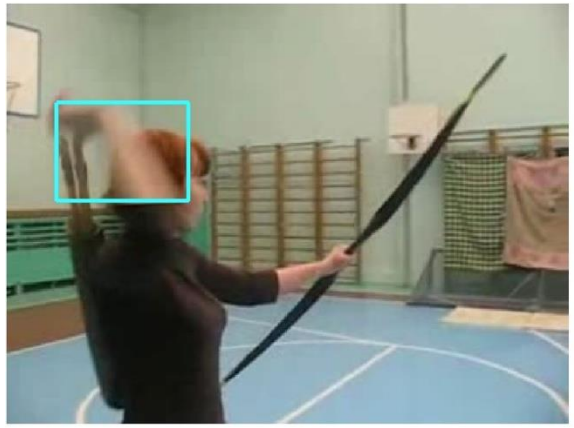
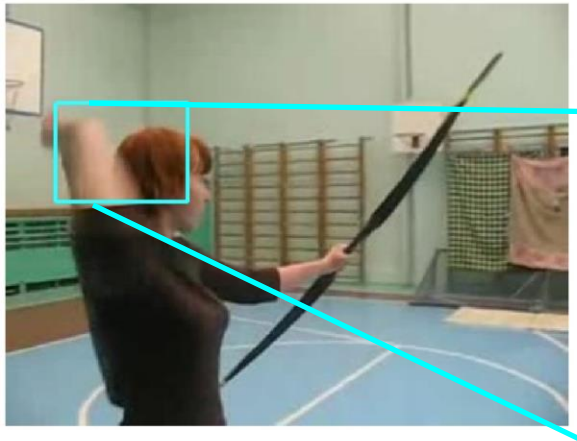


Image at frame t+1

衡量运动（Motion）：光流（Optical Flow）

Image at frame t



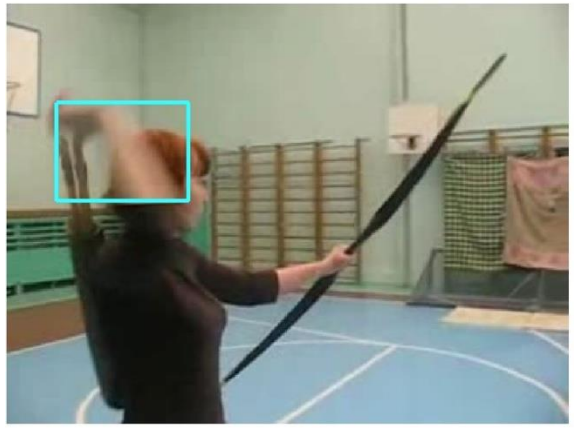
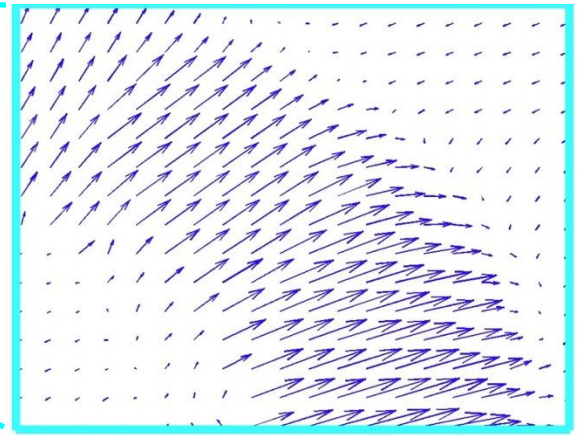


Image at frame t+1

光流在图像 I_t 和 I_{t+1} 之间产生位移场F



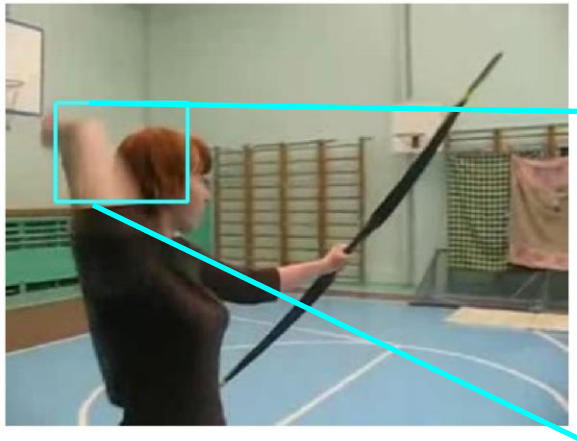
每个像素会在下一帧移动到哪个位置：

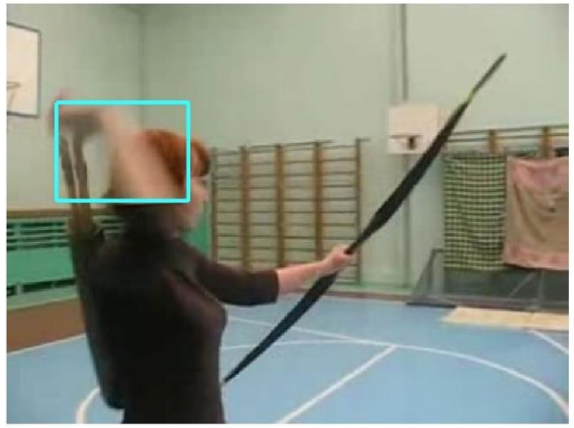
$F(x, y) = (dx, dy)$

$I_{t+1}(x+dx, y+dy) = I_t(x, y)$

- 衡量运动（Motion）：光流（Optical Flow）

Image at frame t





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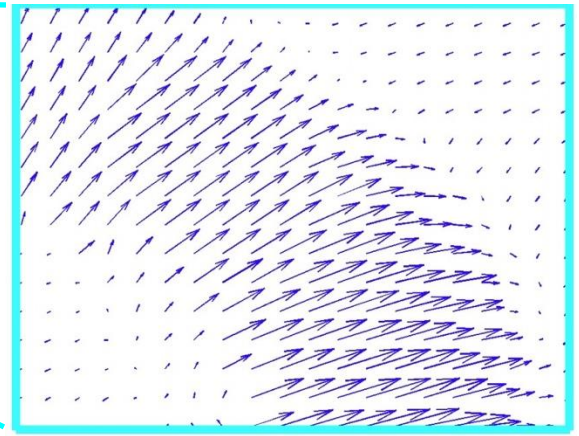


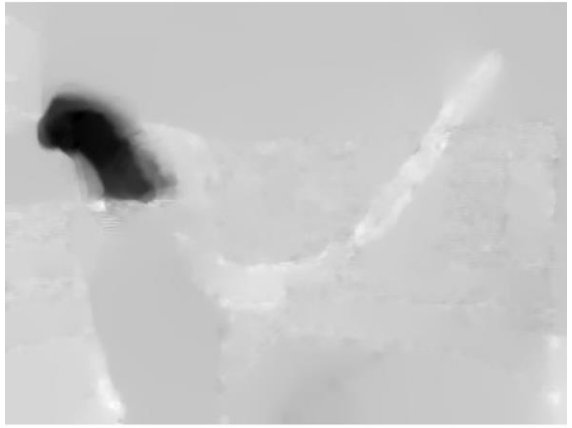
每个像素会在下一帧移动到哪个位置：

$F(x, y) = (dx, dy)$

$I_{t+1}(x+dx, y+dy) = I_t(x, y)$

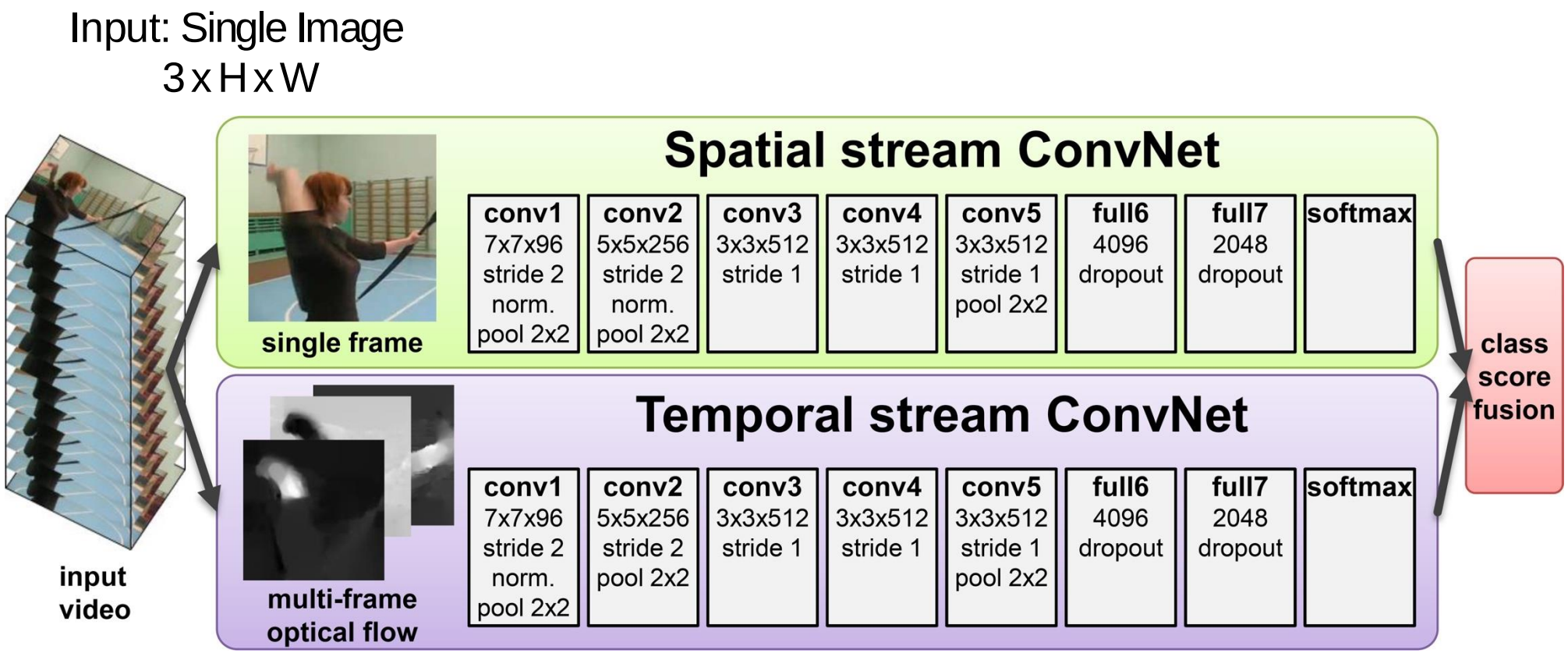
光流可以突出局部运动

Horizontal flow dx

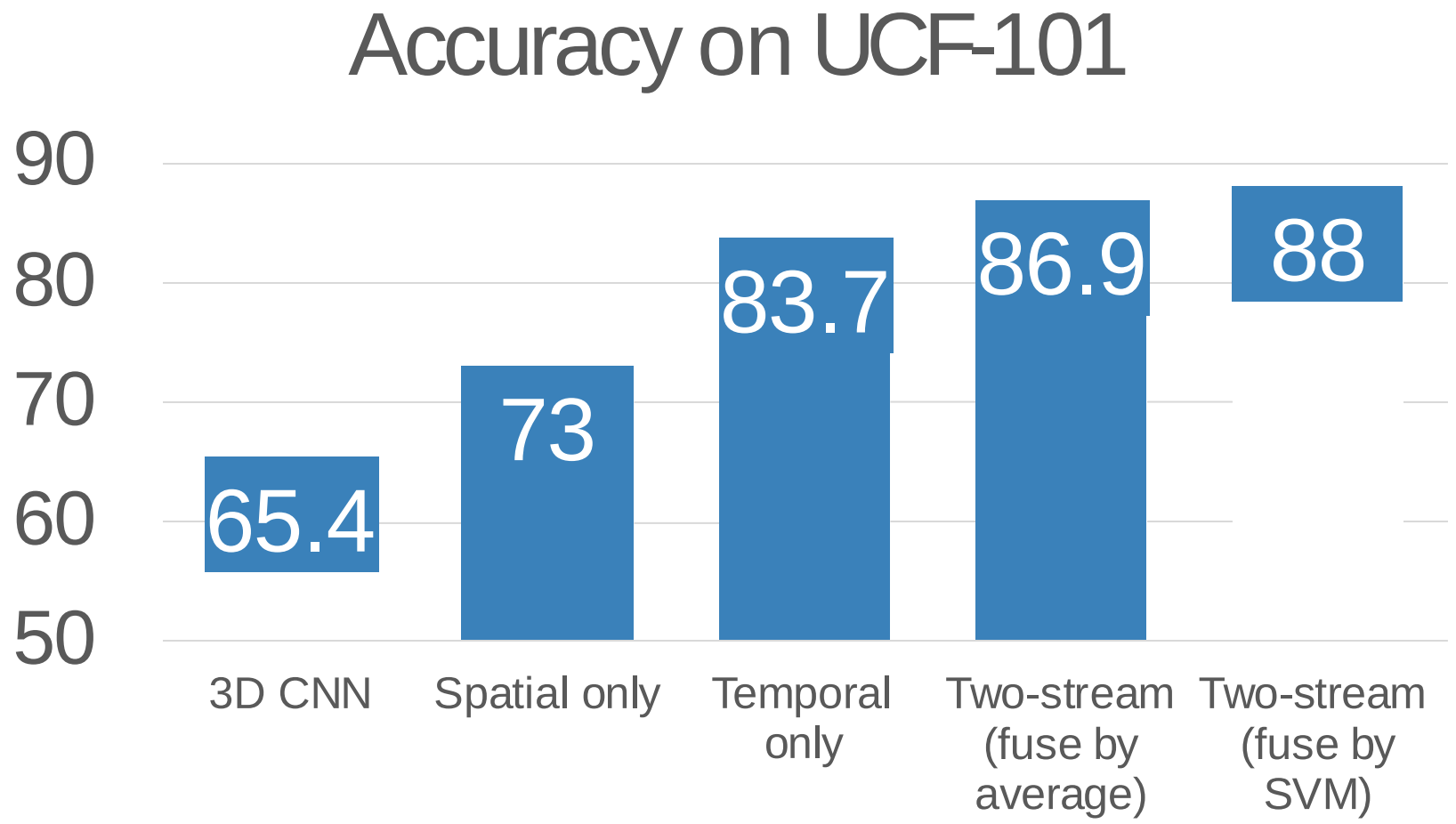


Vertical Flow dy

分别处理 motion 和 appearance：Two-Stream Networks

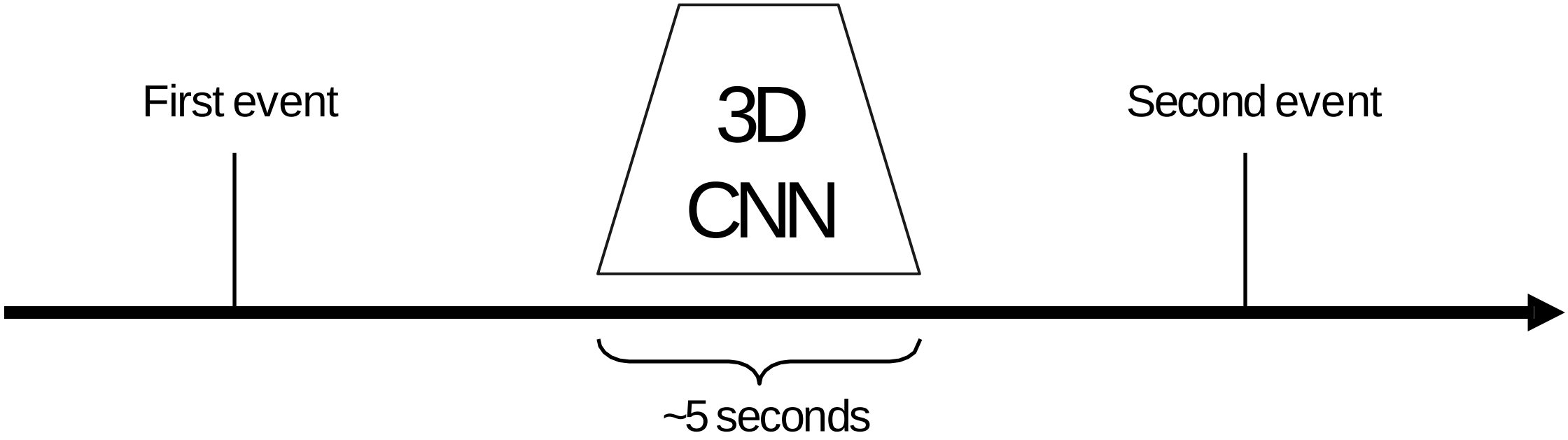


- 分别处理 motion 和 appearance：Two-Stream Networks



视频理解

- 建模长时间时序结构
- 目前为止，我们主要关注约2-5秒的非常短的片段中模拟帧之间的局部运动。如何处理长时间时序结构呢?



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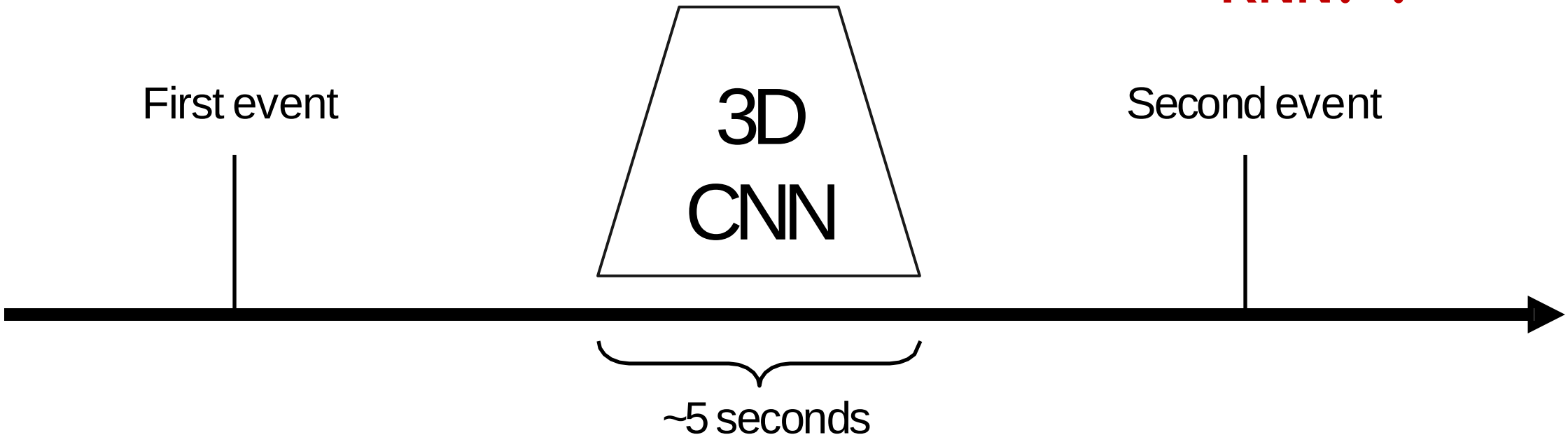
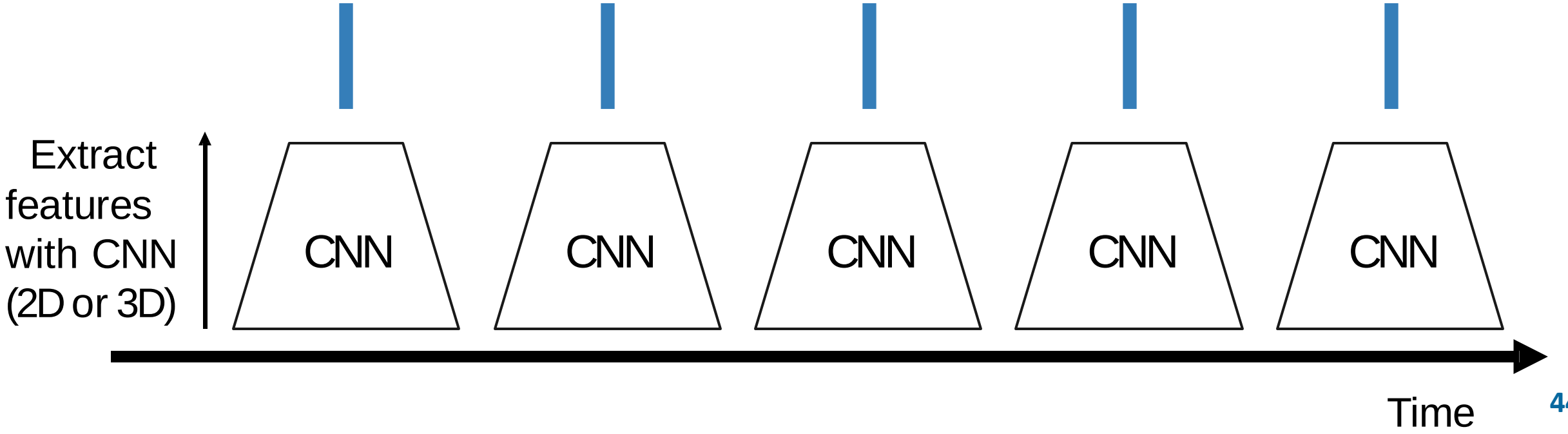
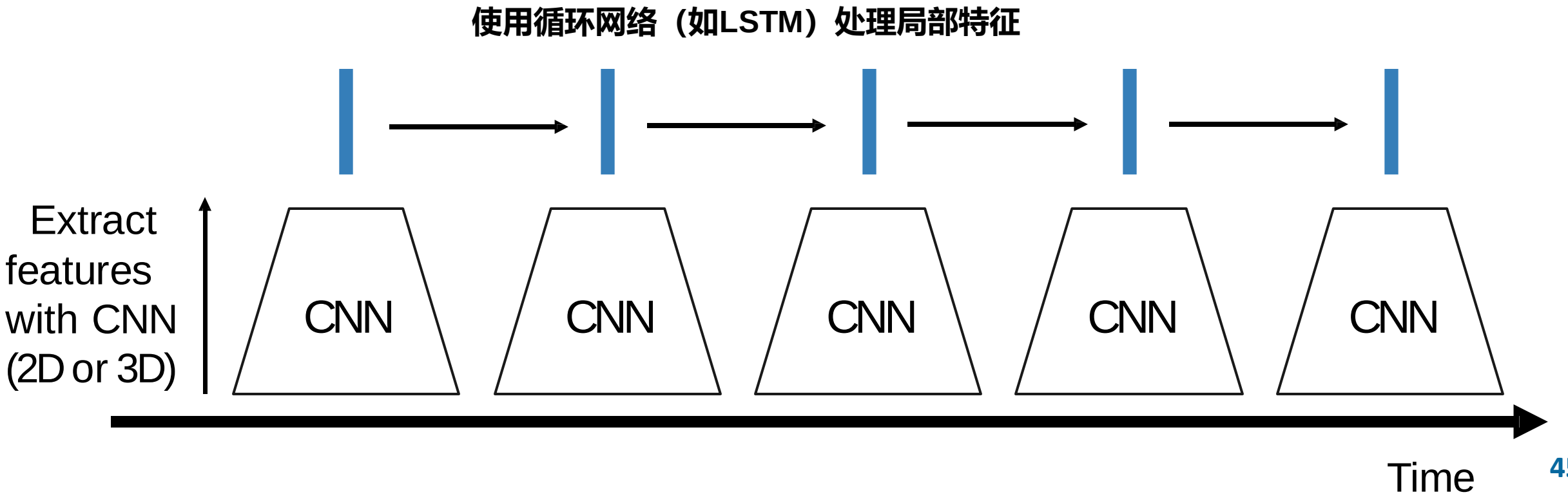
RNN？！



- 建模长时间时序结构

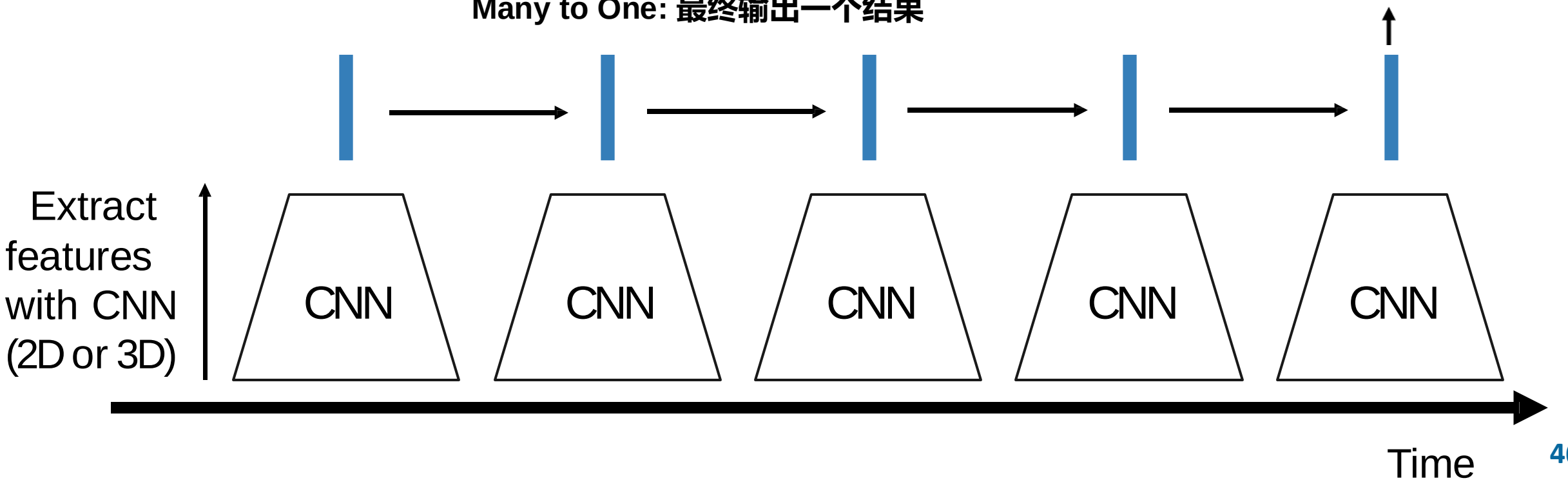


建模长时间时序结构


使用循环网络（如LSTM）处理局部特征
Extract features with CNN (2D or 3D)
CNN
CNN
CNN
CNN
CNN
Time

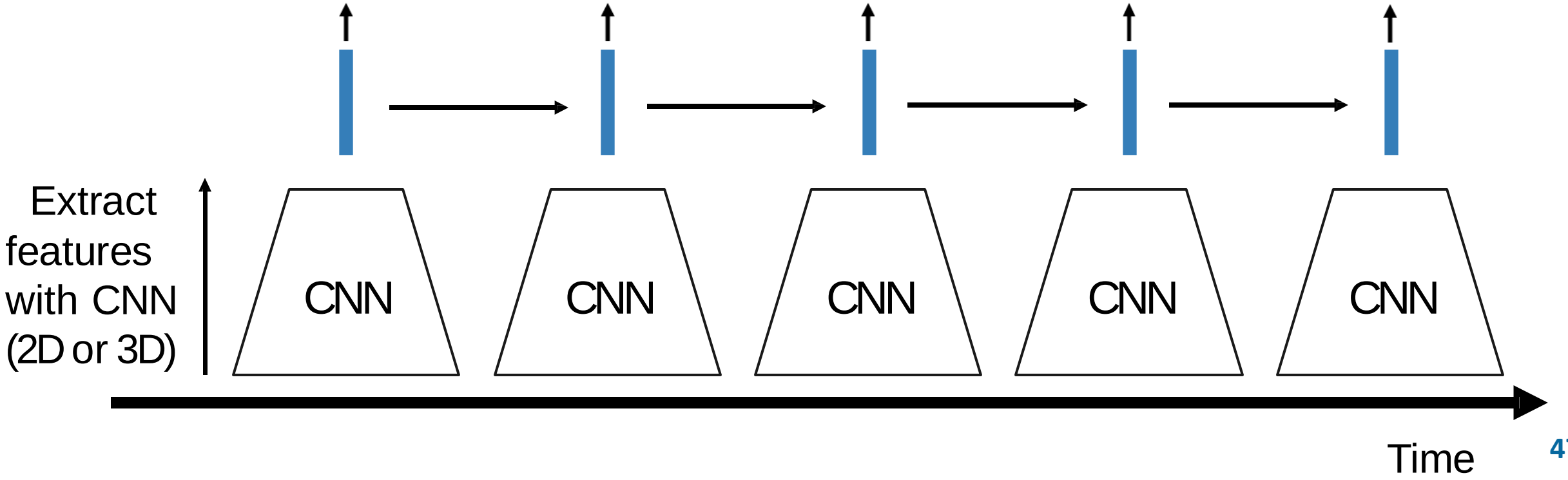
建模长时间时序结构

使用循环网络（如LSTM）处理局部特征
Many to One: 最终输出一个结果

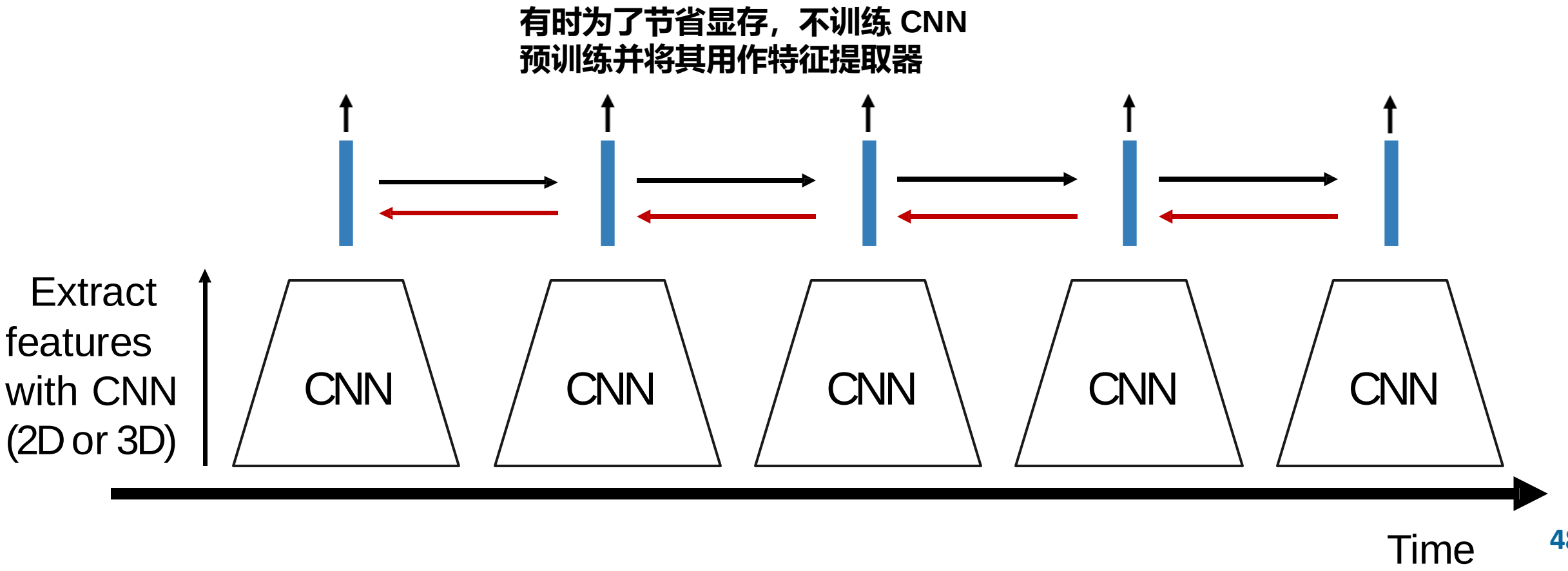


建模长时间时序结构

使用循环网络（如LSTM）处理局部特征
Many to Many: 为每一帧输出结果

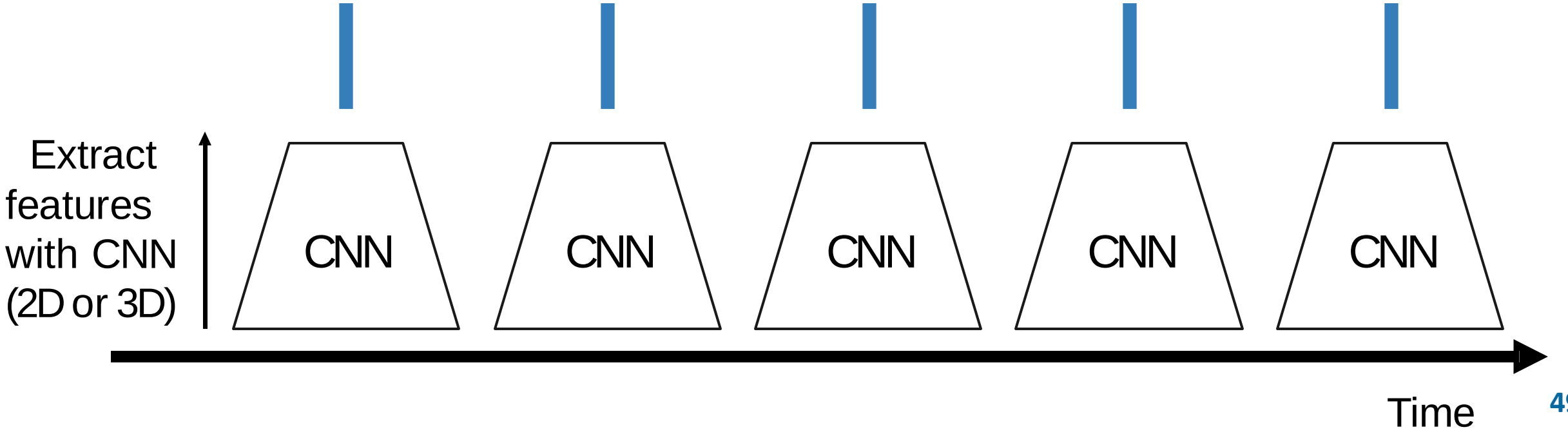


建模长时间时序结构


有时为了节省显存，不训练 CNN
预训练并将其用作特征提取器
Extract features with CNN (2D or 3D)
CNN
CNN
CNN
CNN
CNN
Time

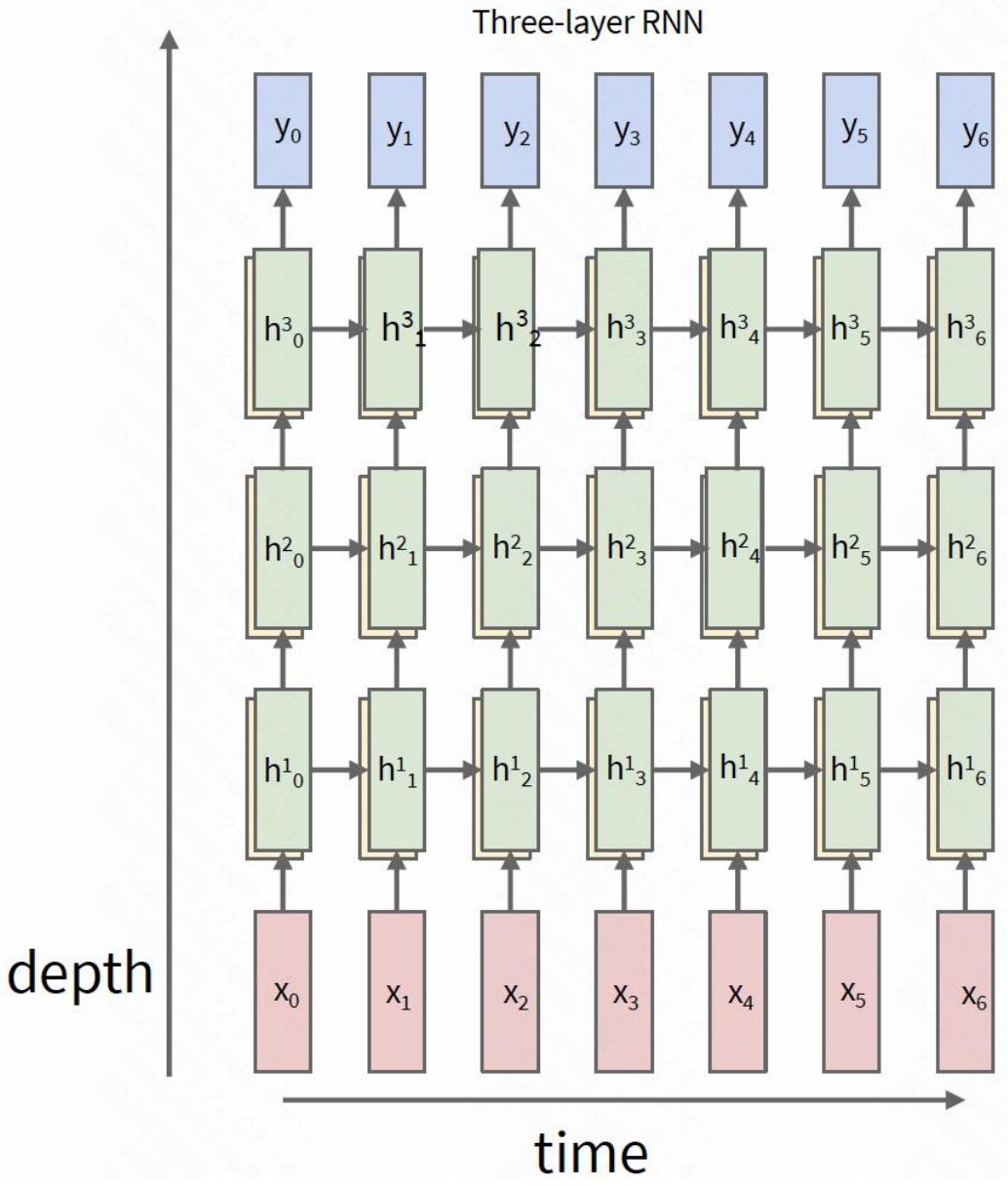
建模长时间时序结构

CNN内部：每个特征值都是固定时间窗口的函数（局部时间结构）
RNN内部：每个向量都是所有先前向量的函数（全局时间结构）
我们可以结合这两种方法吗?

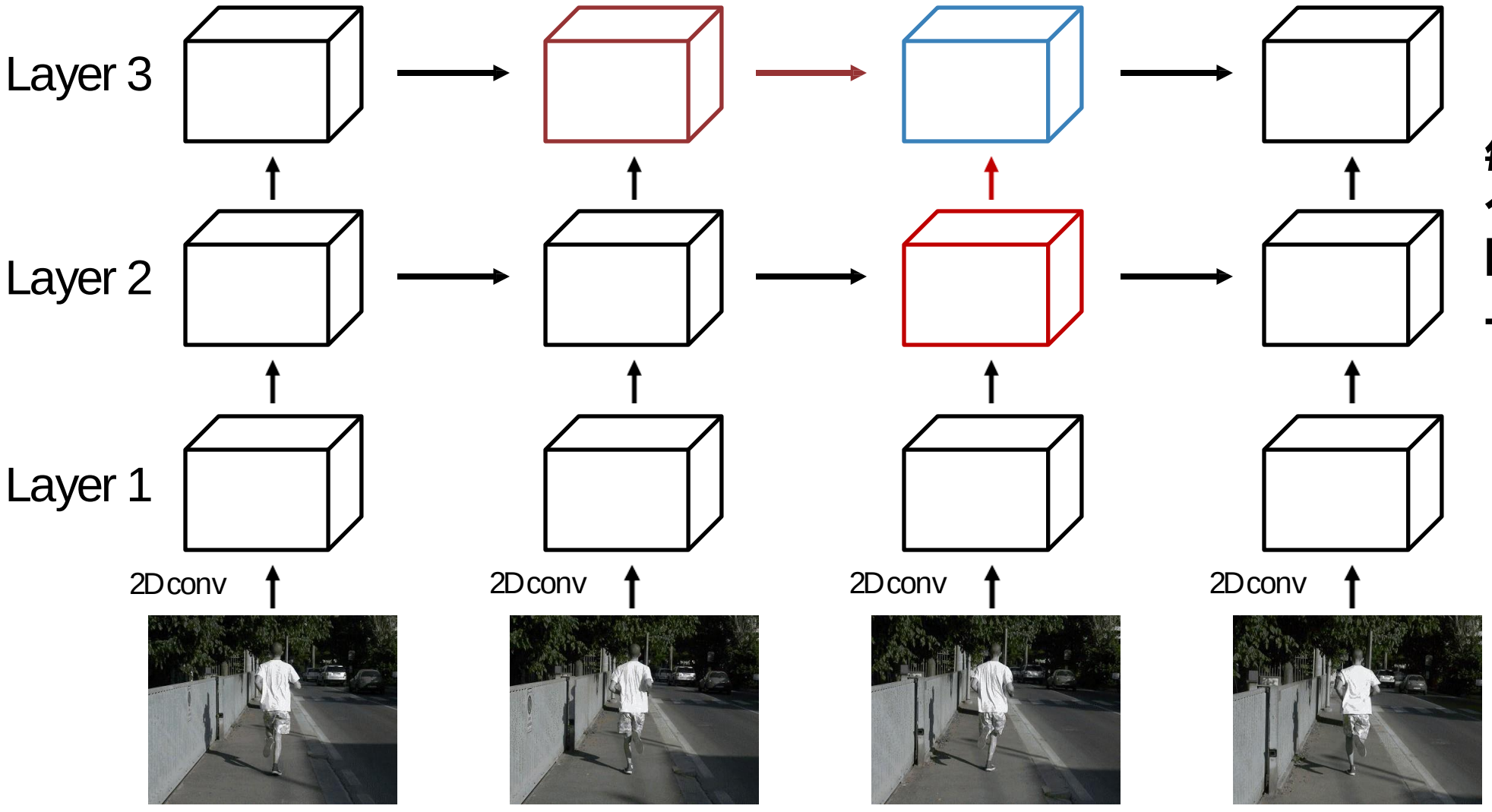


视频理解

- 回顾：多层 RNN
  - 我们可以使用类似的结构来处理视频



视频理解



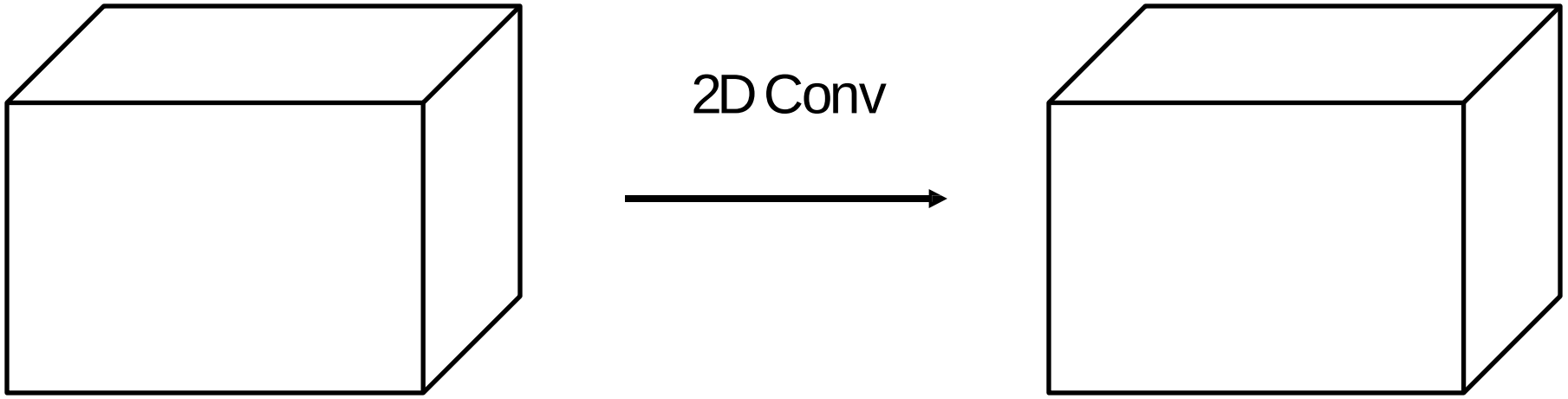
整个网络处理二维特征图：
C x H x W

每个输出都取决于两个输入：
同一层，前一时间步
上一层，相同时间步

在每一层使用不同的权重，跨时间共享权重

Recurrent Convolutional Network

Normal 2D CNN:



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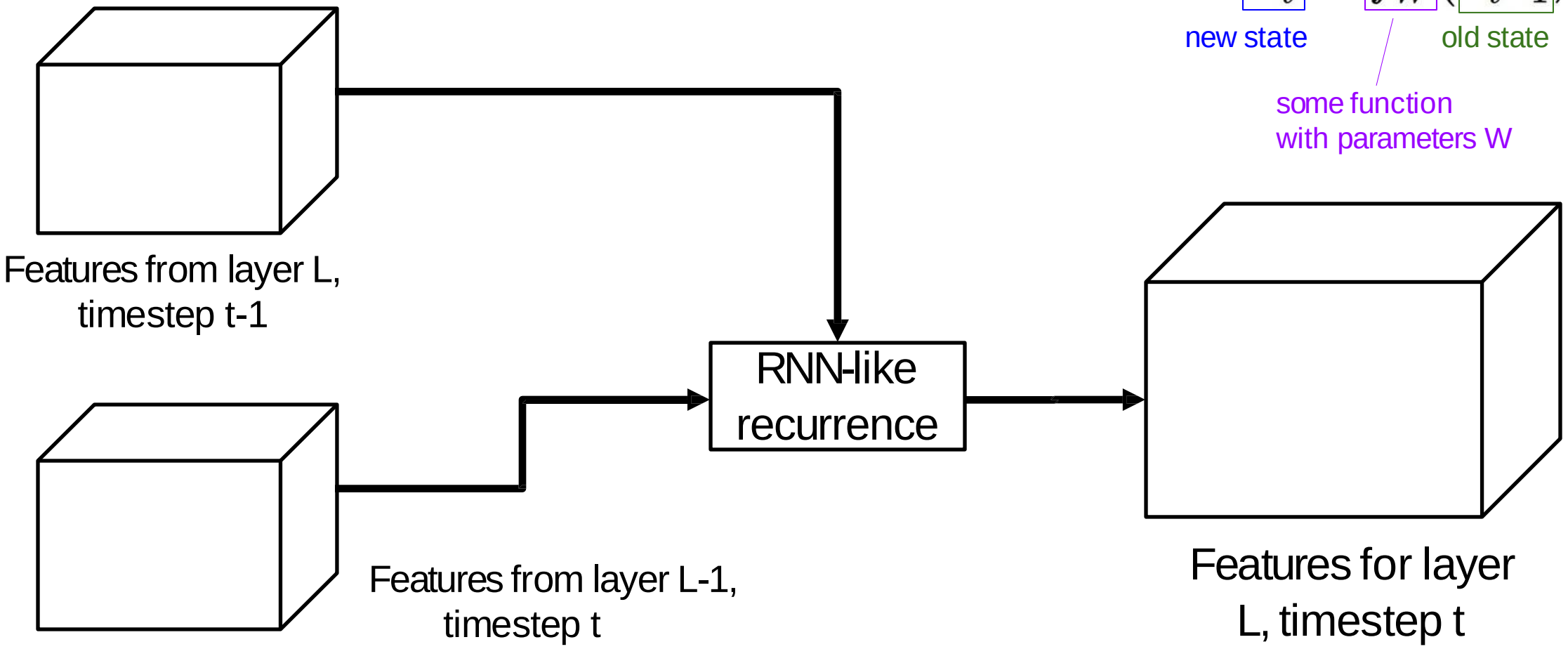
Recurrent Convolutional Network

回顾：RNN

$$h_t = f_W(h_{t-1}, x_t)$$

new state　　old state

some function with parameters W



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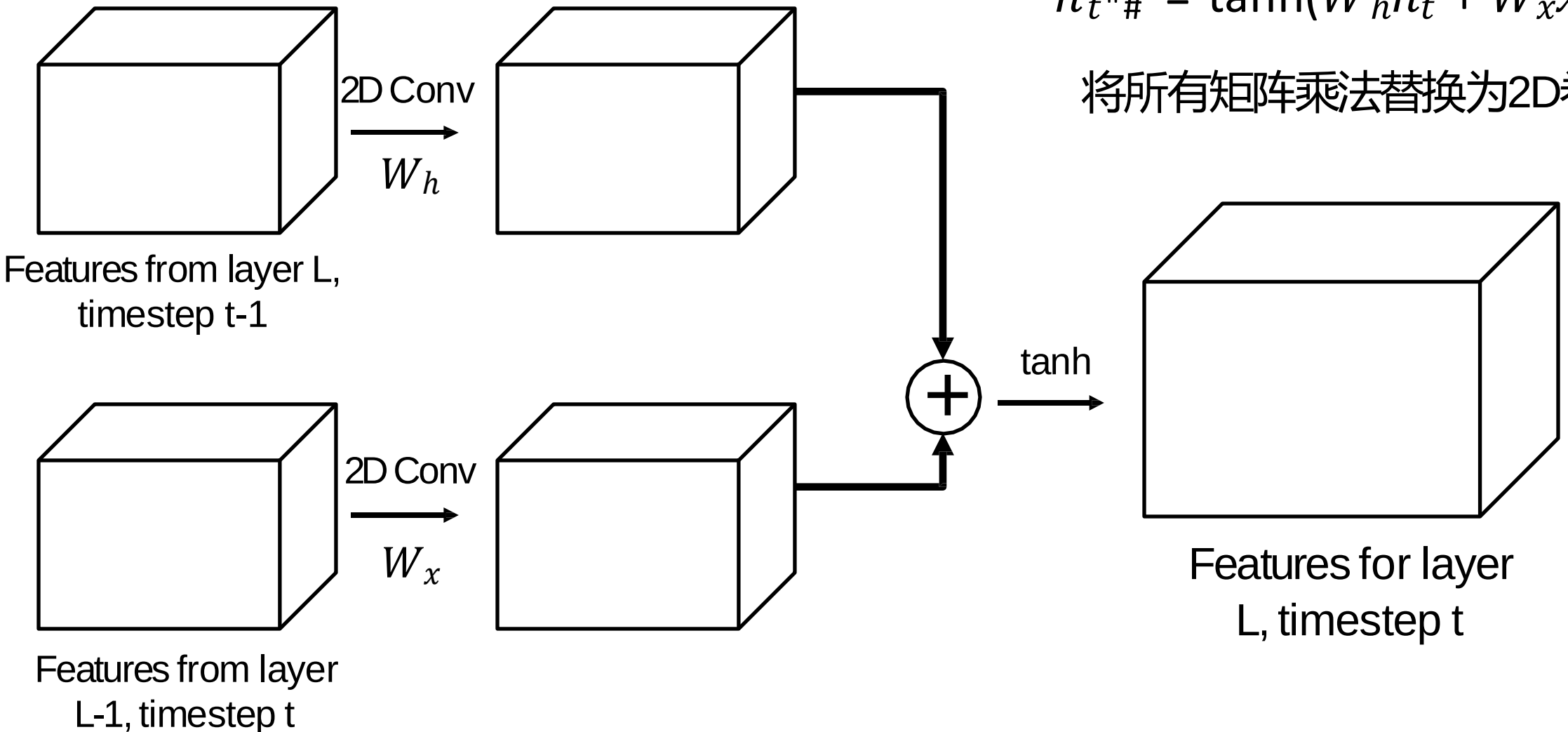
Recurrent Convolutional Network



回顾: Vanilla RNN

$$h_t = \tanh(W_h h_t + W_x x)$$

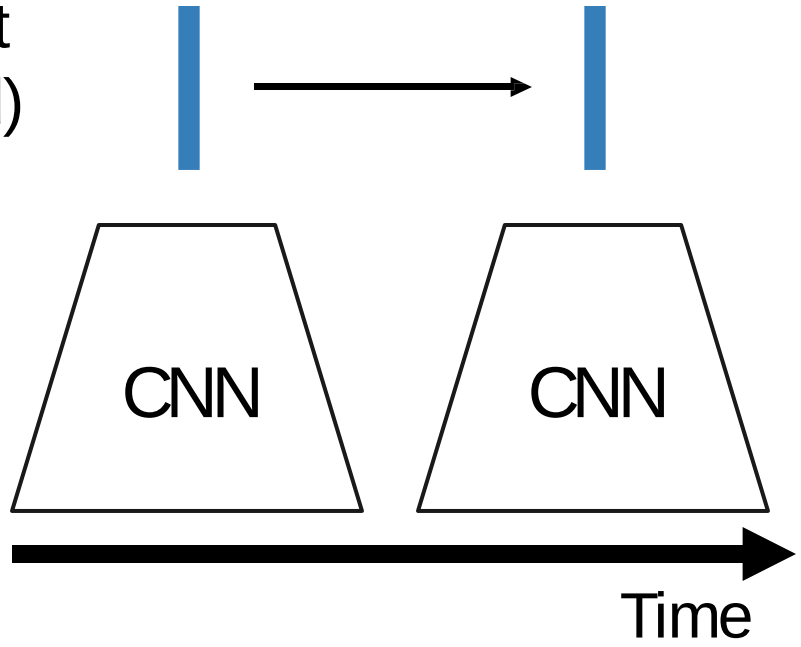
将所有矩阵乘法替换为2D卷积

建模长时间时序结构

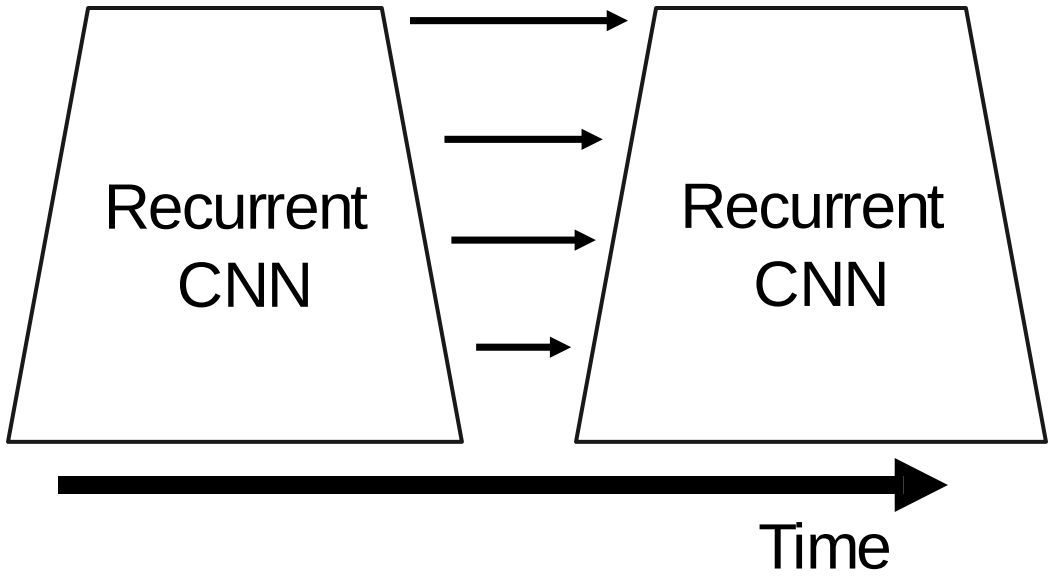
问题：RNN对于长序列很慢（无法并行化）

RNN: Infinite temporal extent (fully-connected)

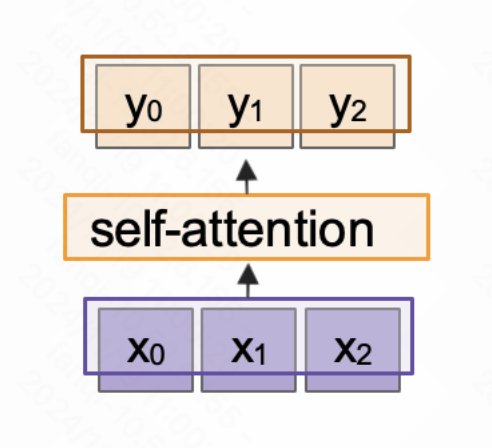
CNN: finite temporal extent (convolutional)

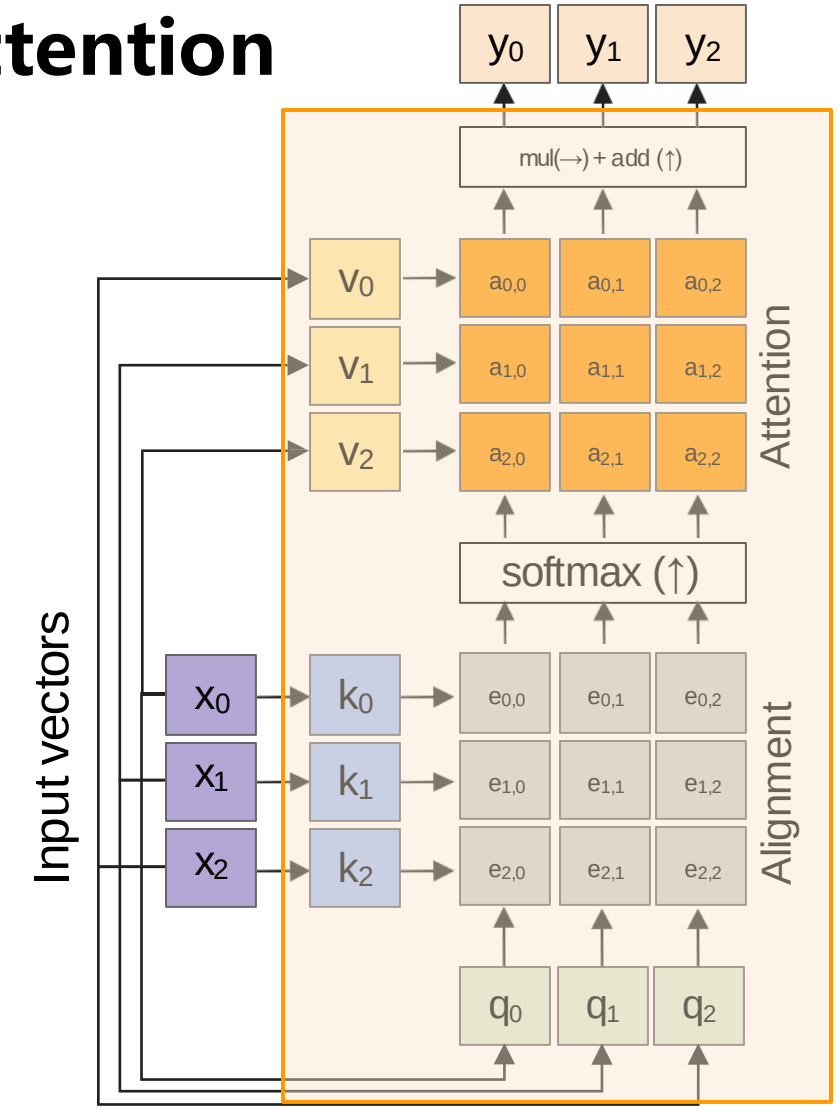


Recurrent CNN: Infinite temporal extent (convolutional)



回顾：Self-Attention



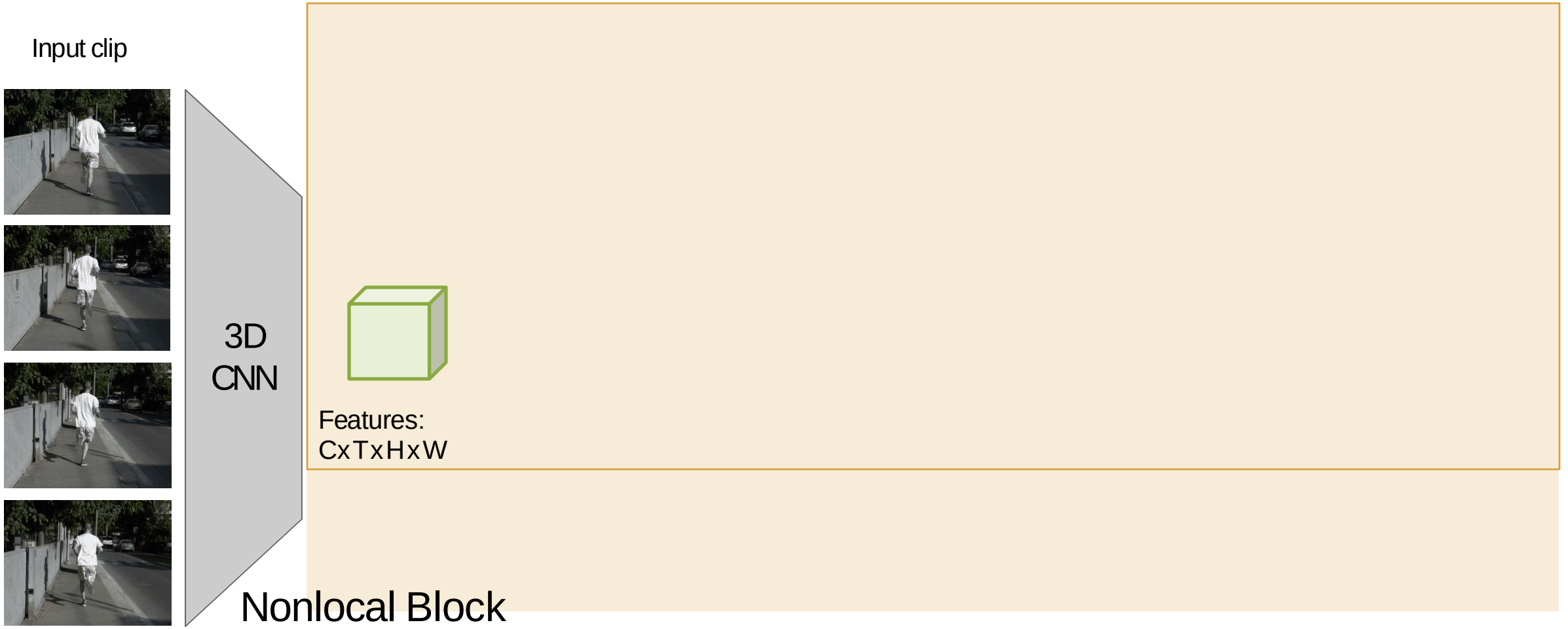


Outputs:
context vectors: y (shape: D_v)

Operations:
Key vectors: $k = xW_k$
Value vectors: $v = xW_v$
Query vectors: $q = xW_q$
Alignment: $e_{i,j} = q_j \cdot k_i / \sqrt{D}$
Attention: $a = \text{softmax}(e)$
Output: $y_j = \sum_j a_{i,j} v_i$

Inputs:
Input vectors: x (shape: N x D)

Spatio-Temporal Self-Attention (Nonlocal Block)



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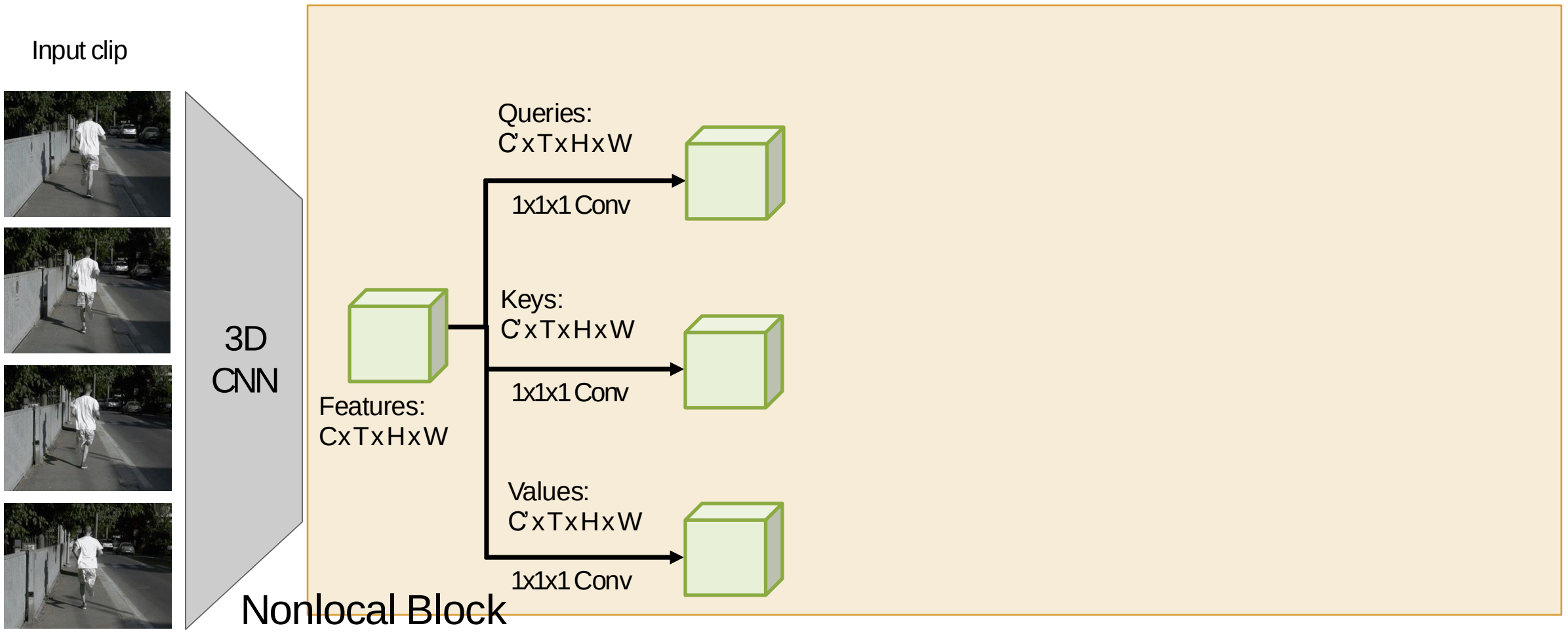
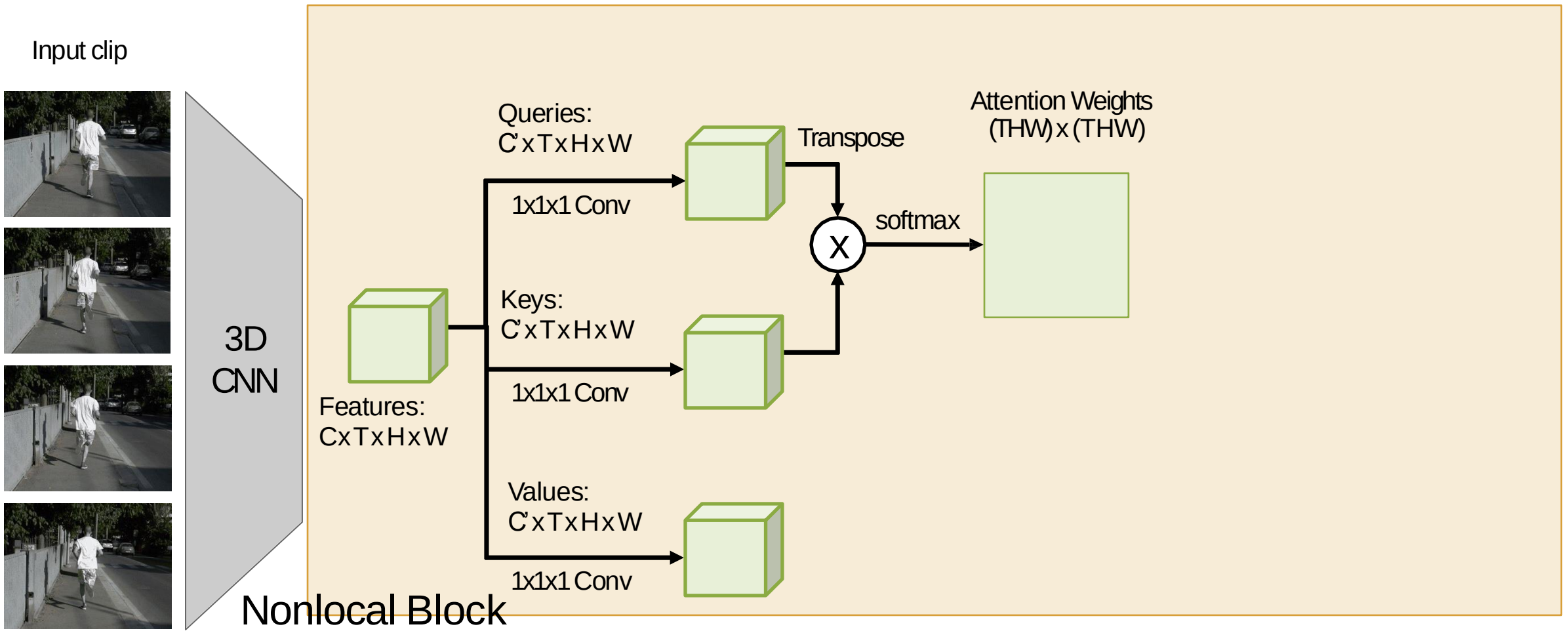


Spatio-Temporal Self-Attention (Nonlocal Block)



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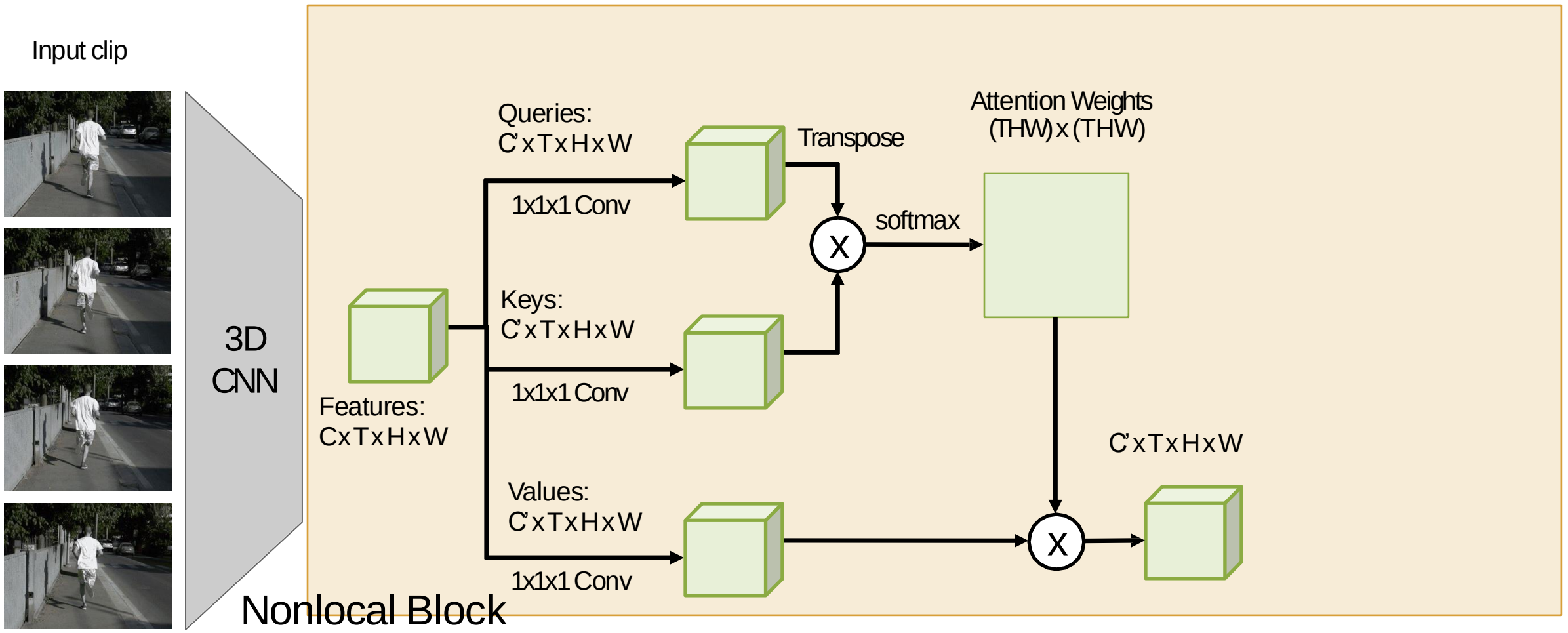

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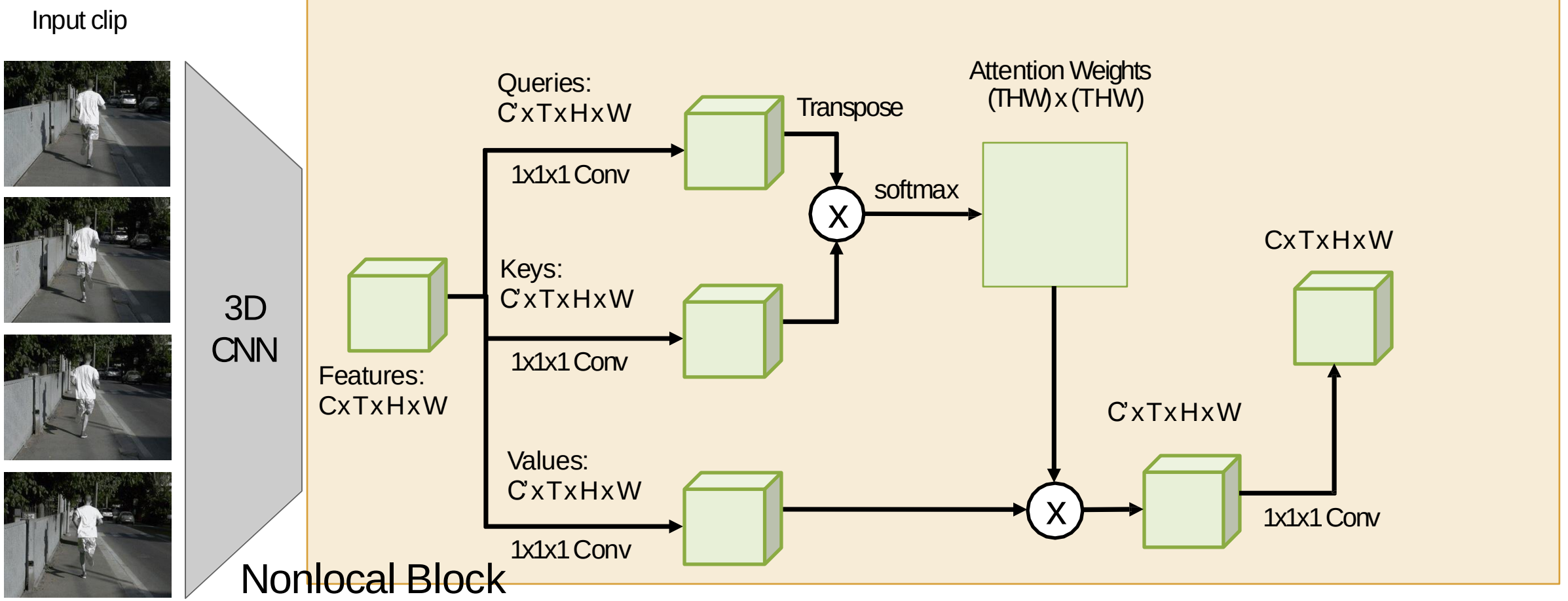

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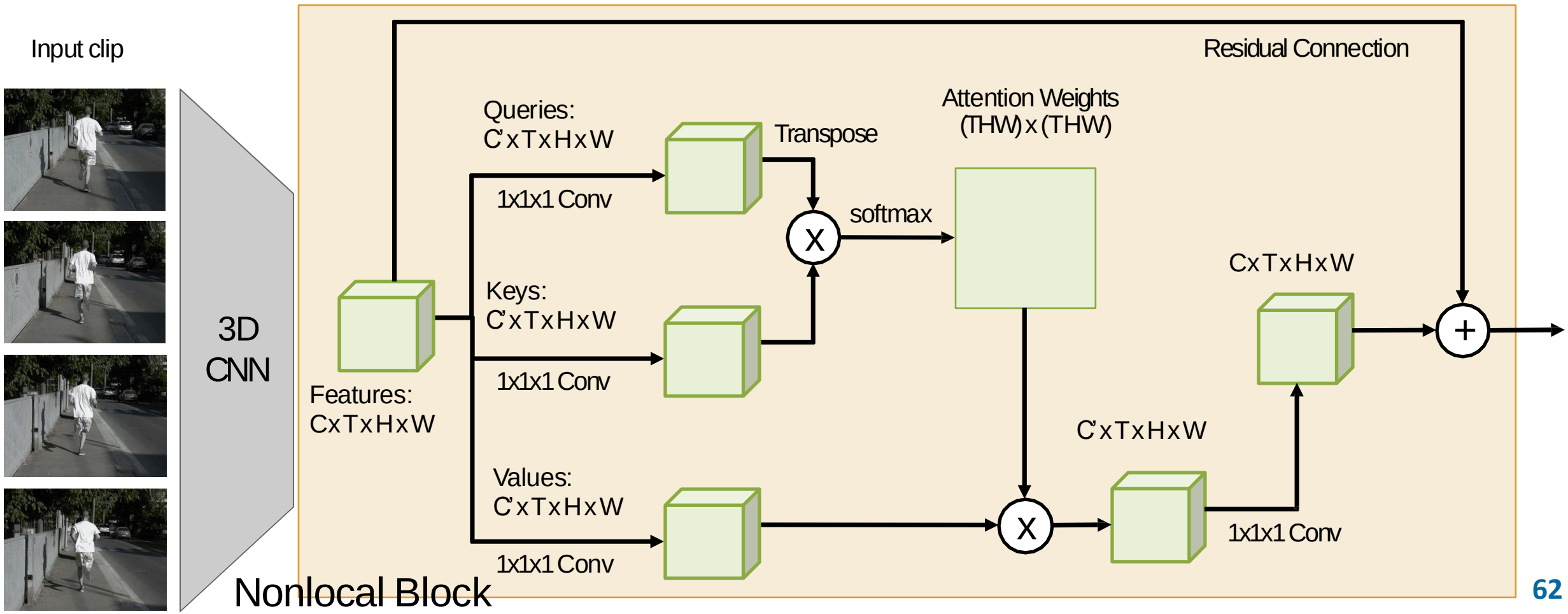
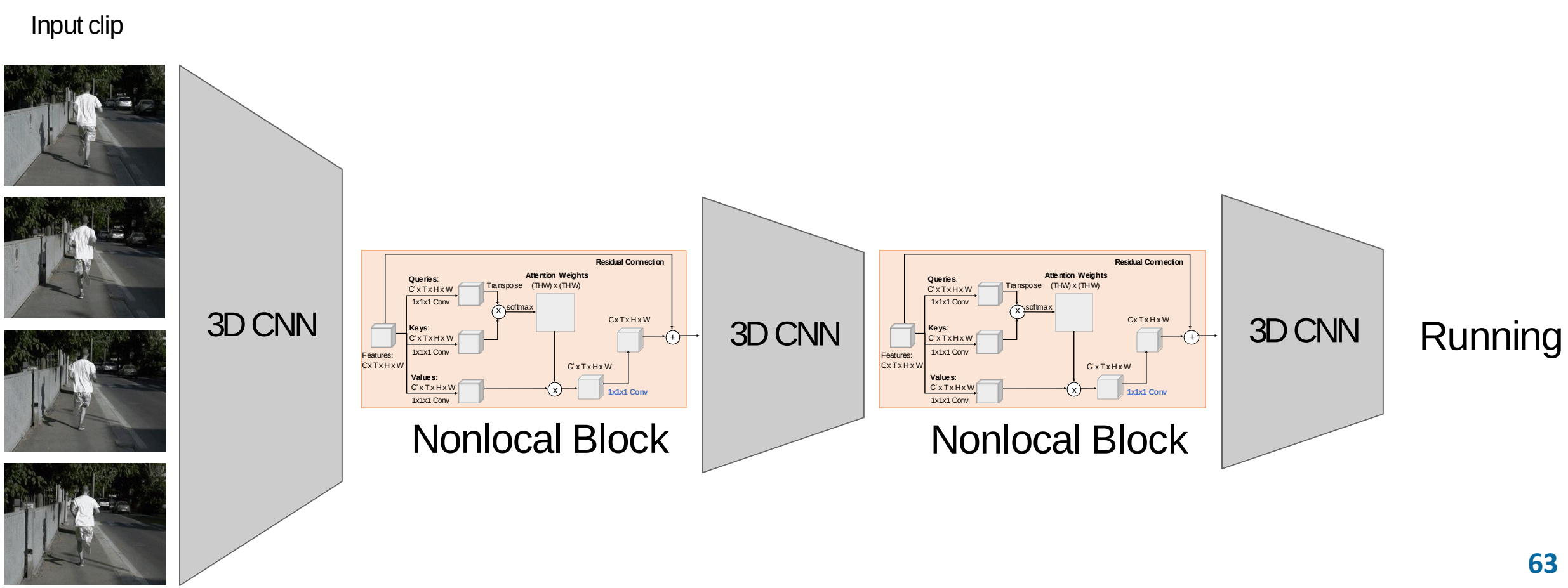


Nonlocal Block

Spatio-Temporal Self-Attention (Nonlocal Block)



视频理解

- 重用 2D CNN：Inflating 2D Networks to 3D (I3D)
- 将每个 2D kernel 替换为 3D kernel

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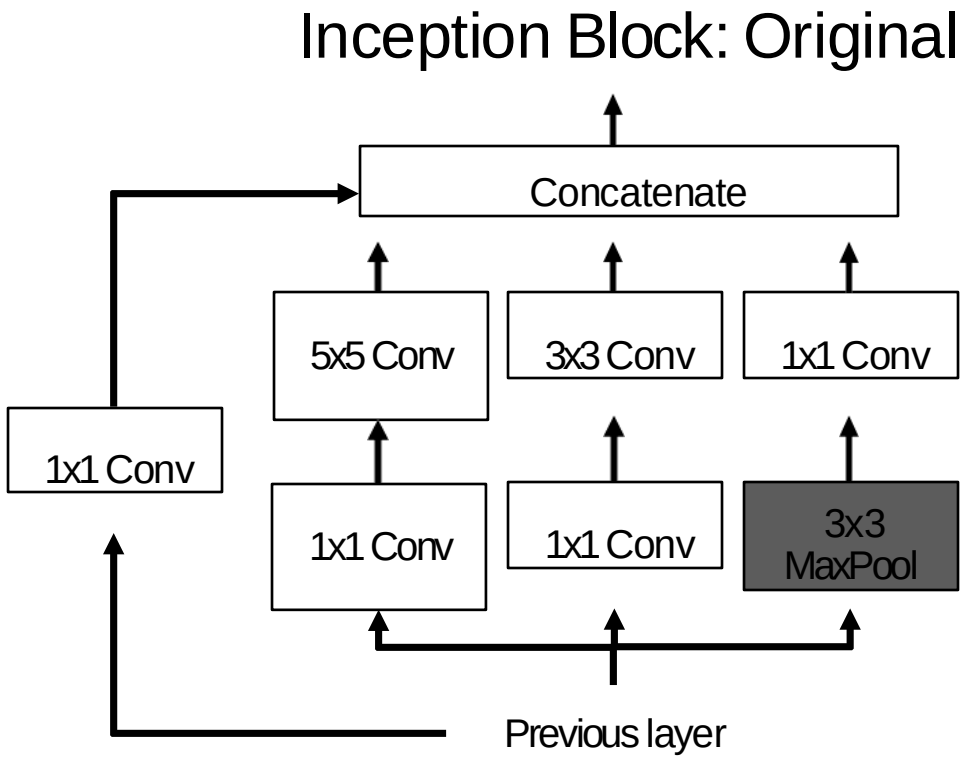

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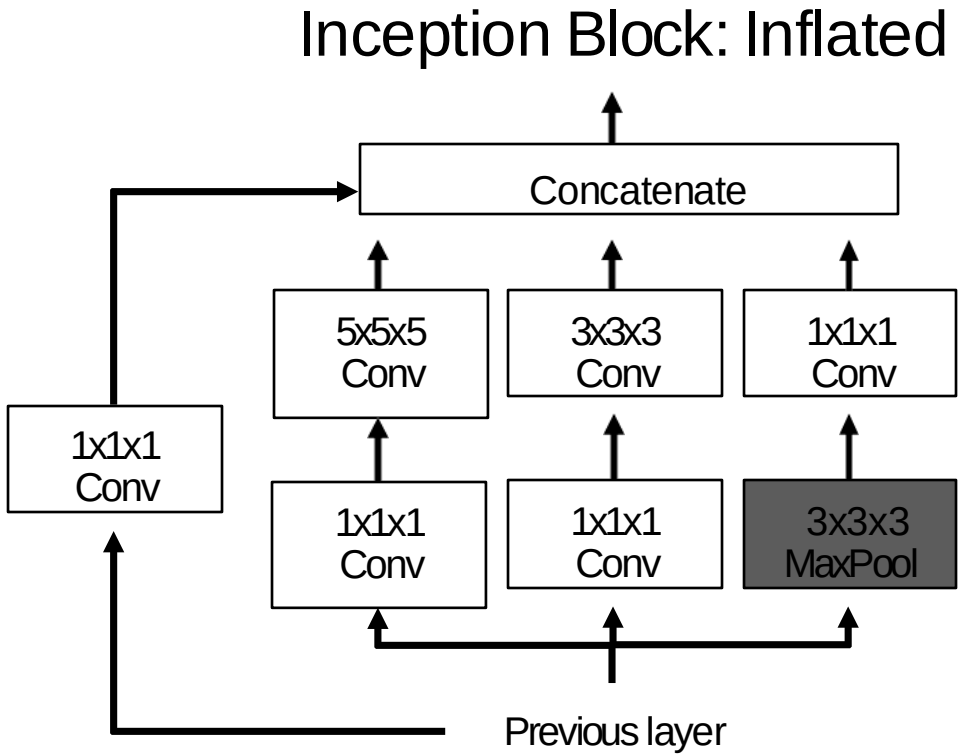
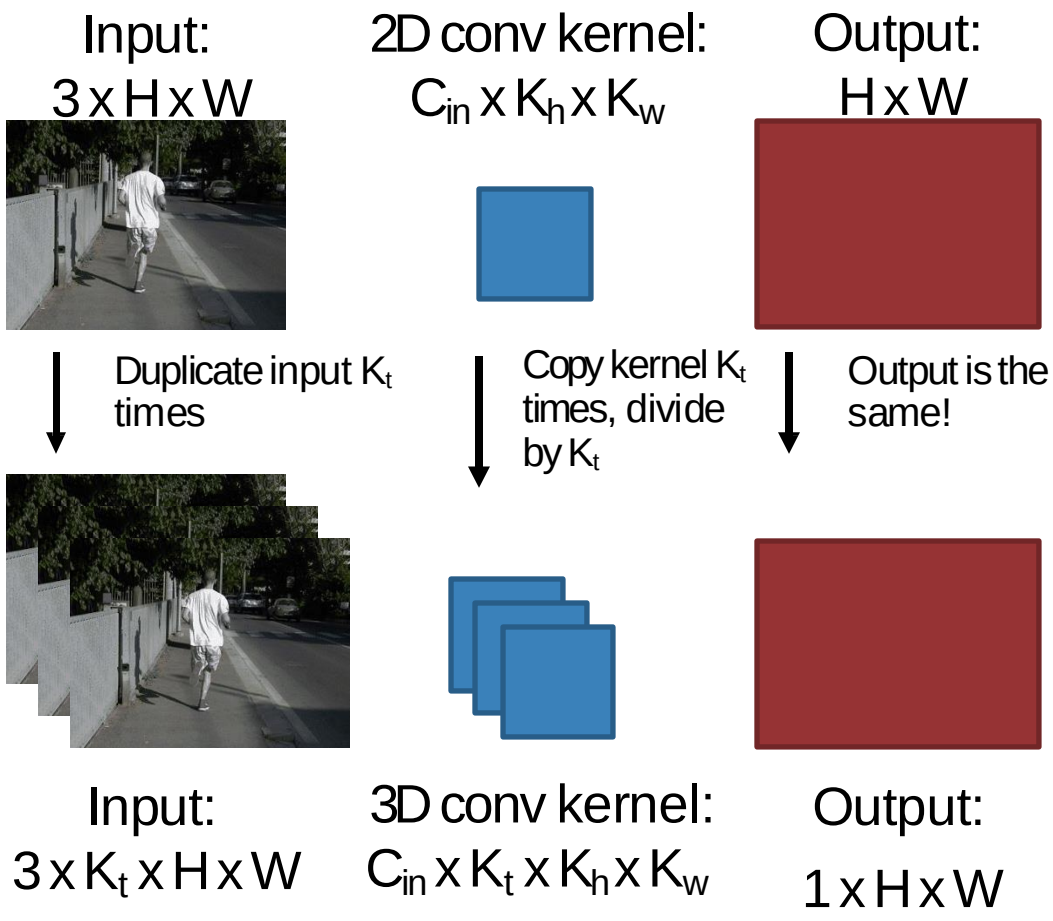


- Inflating 2D Networks to 3D (I3D)
- 可以使用 2D conv 的权重来初始化 3D conv：在空间中复制 Kt 次并除以 Kt
- 这使得两者的预测结果相同



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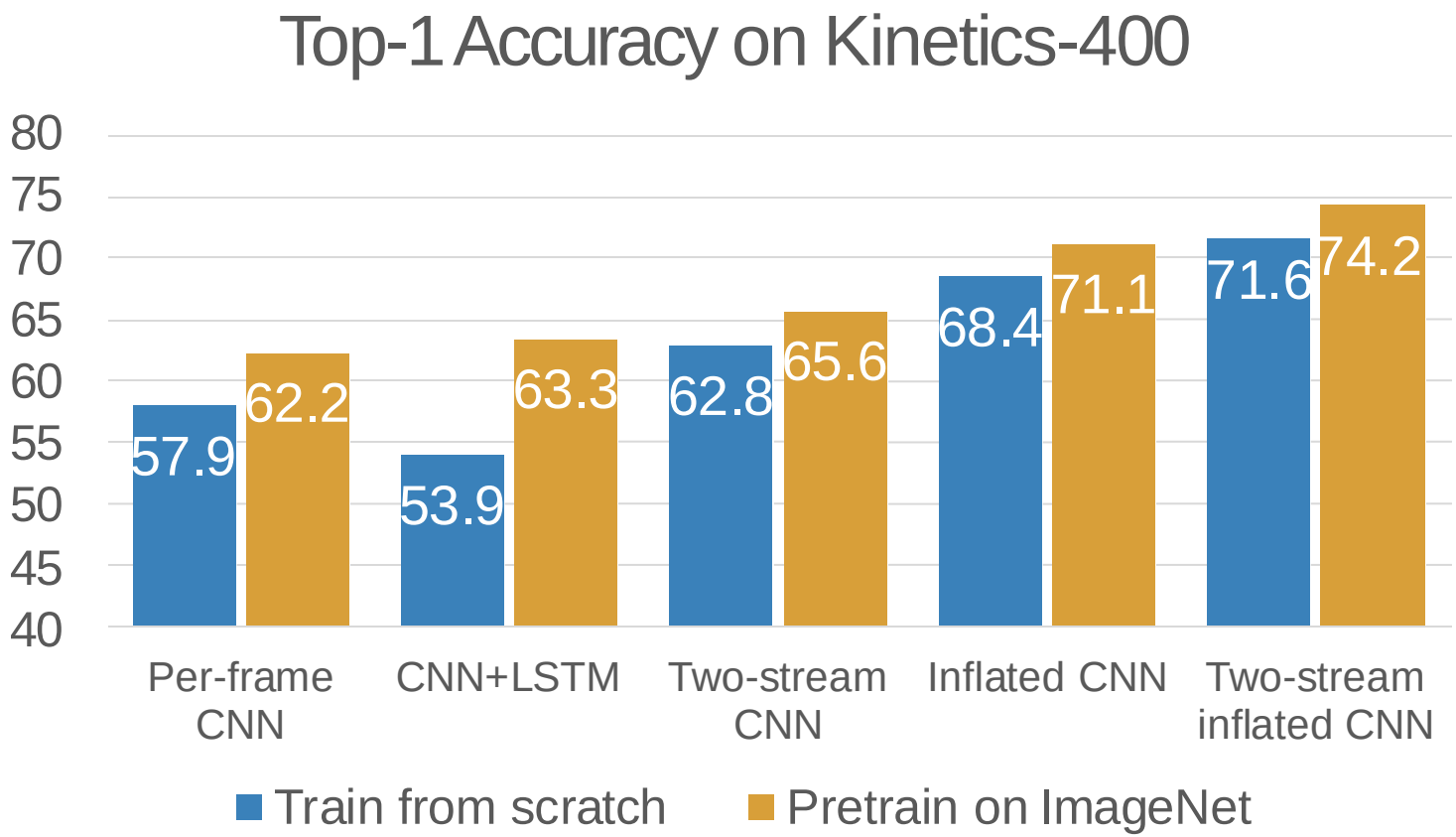
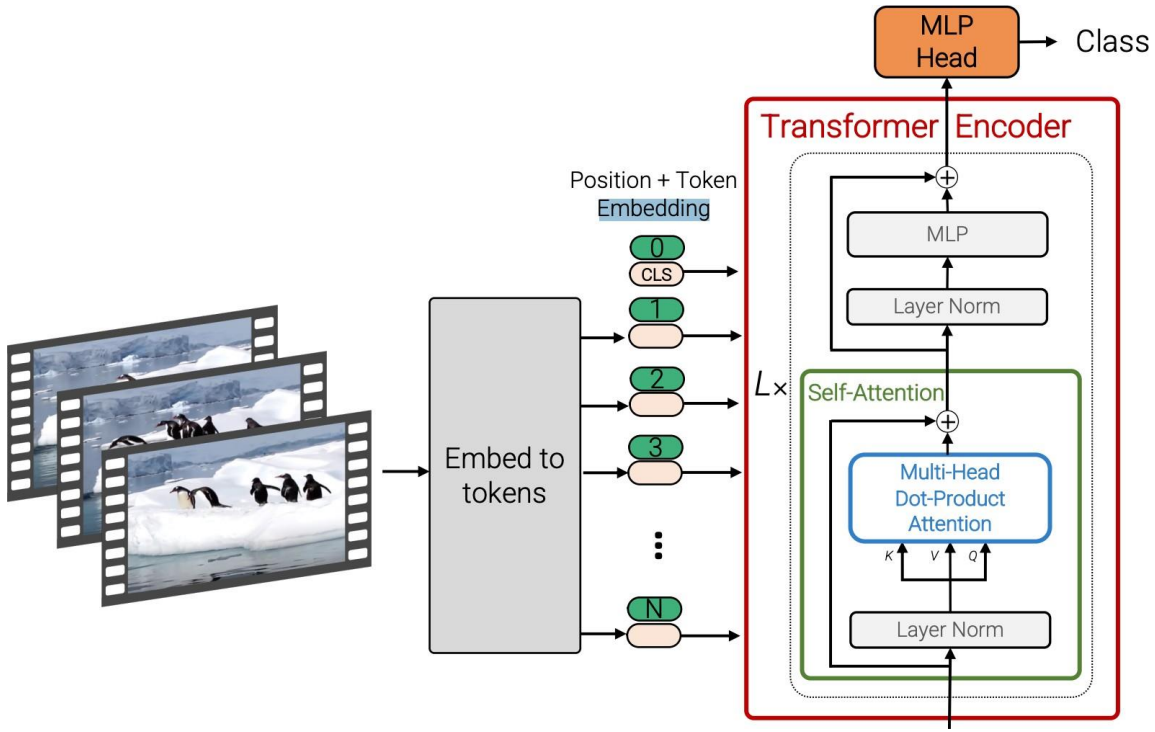
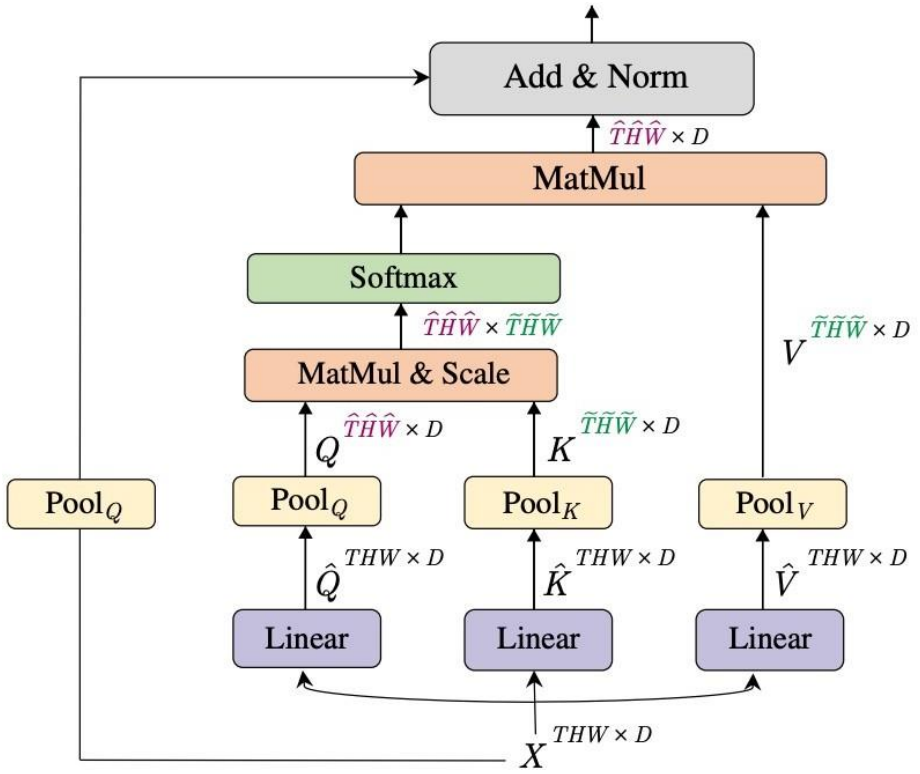


Video Transformer





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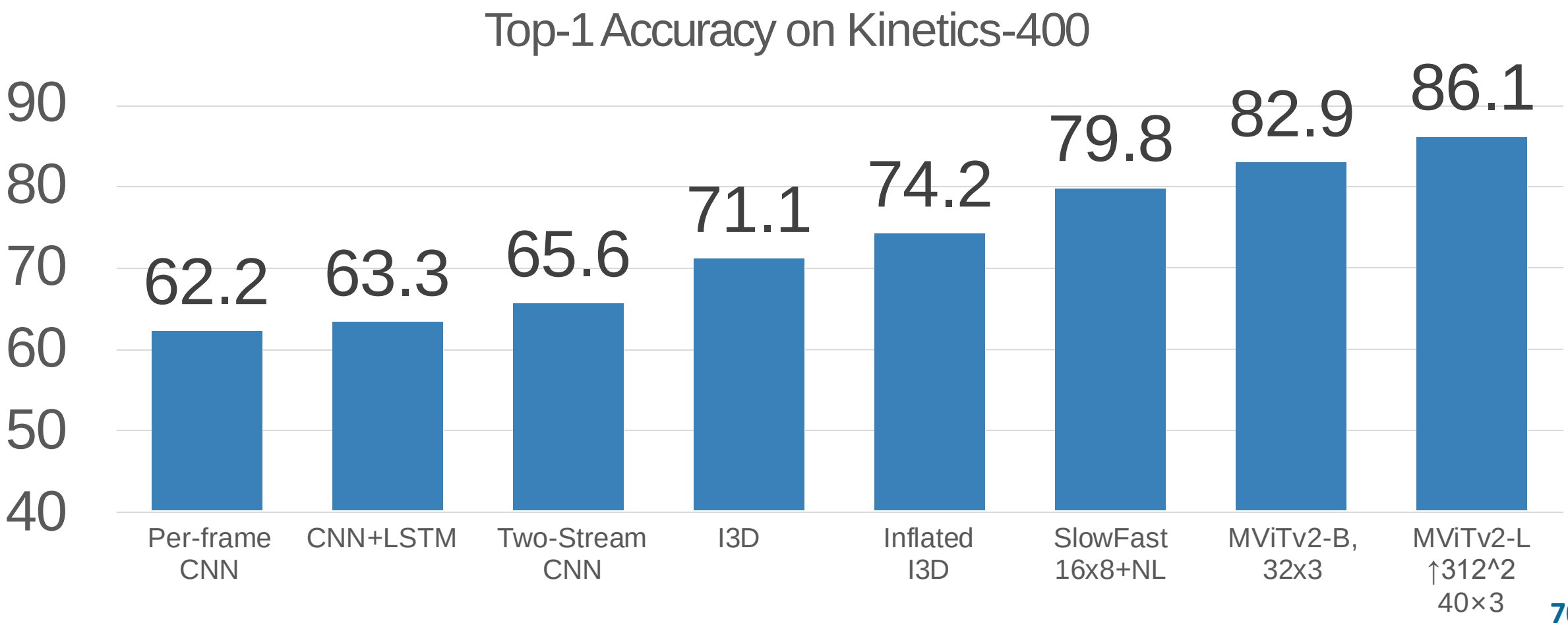
Video Transformer



视频理解

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视频任务

视频分类：识别动作

Swimming
Running
Jumping
Eating
Standing

视频理解

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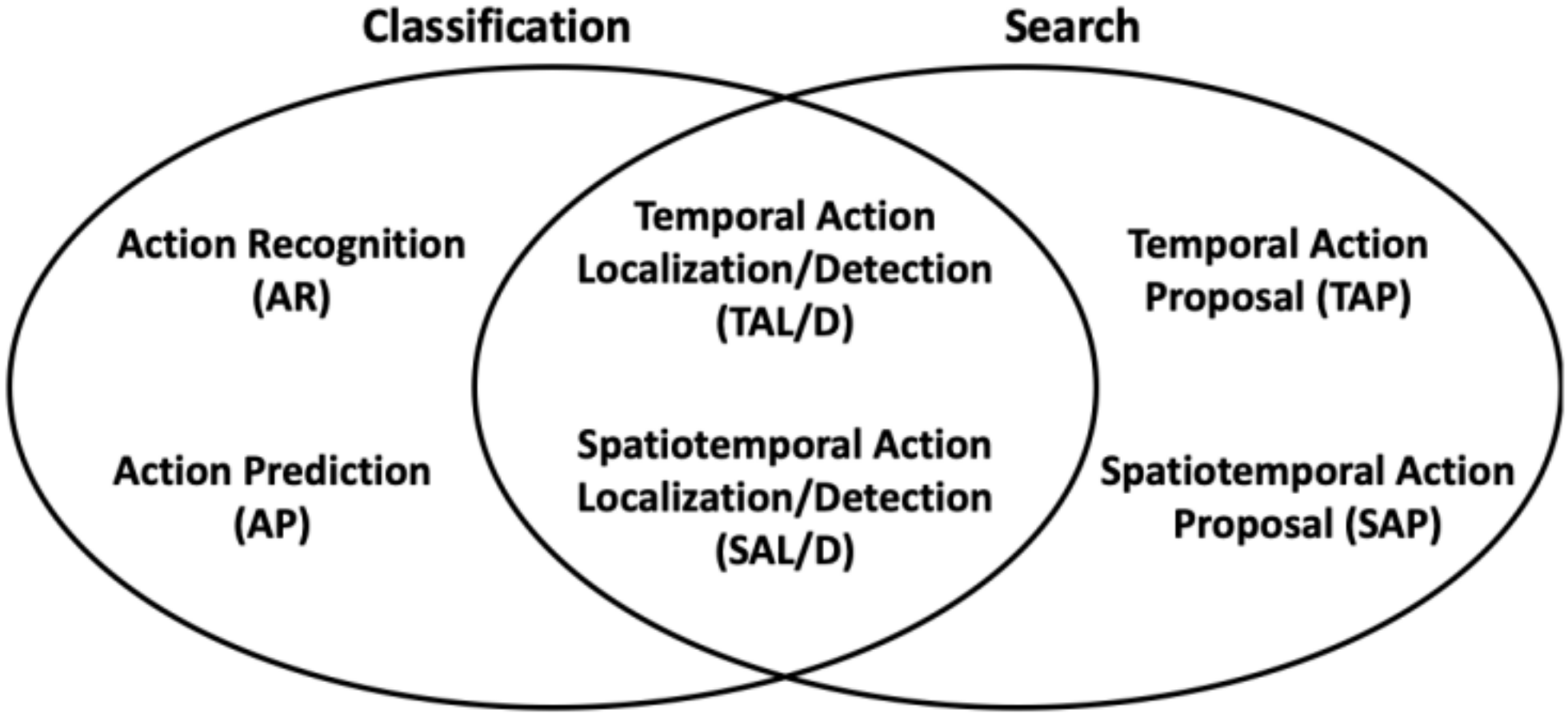
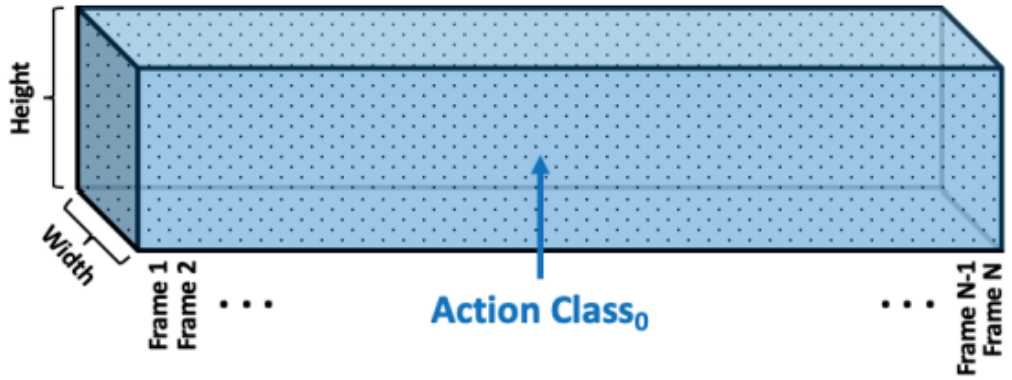
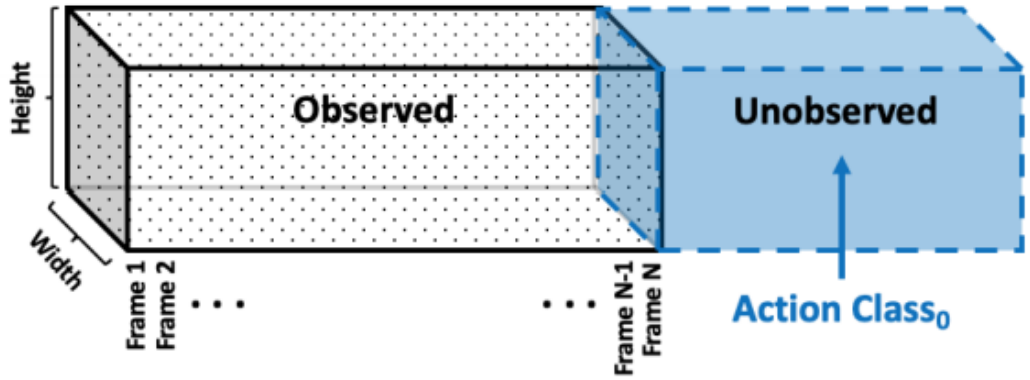
视频任务



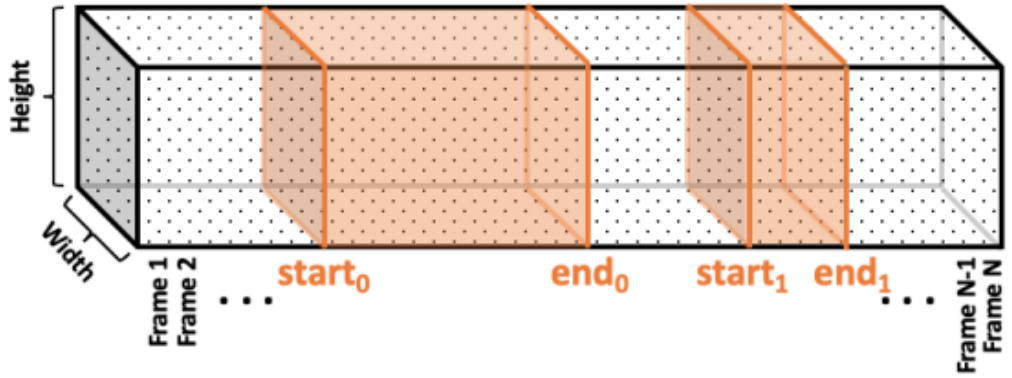
Action Recognition (AR)



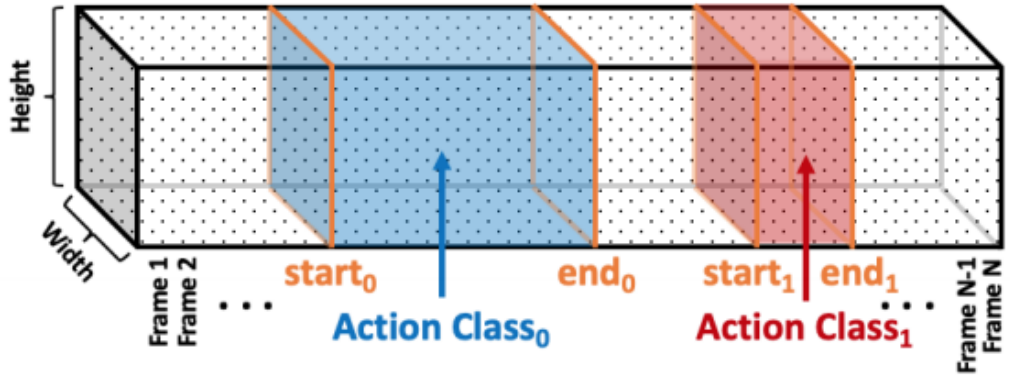
Action Prediction (AP)



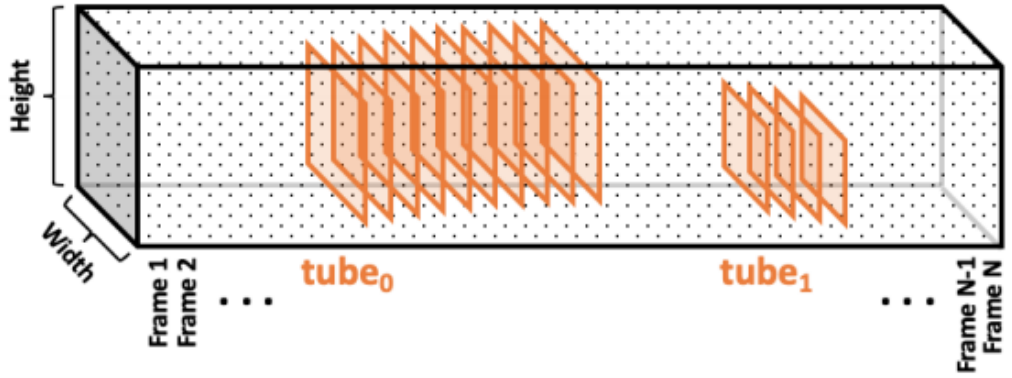
Temporal Action Proposal (TAP)



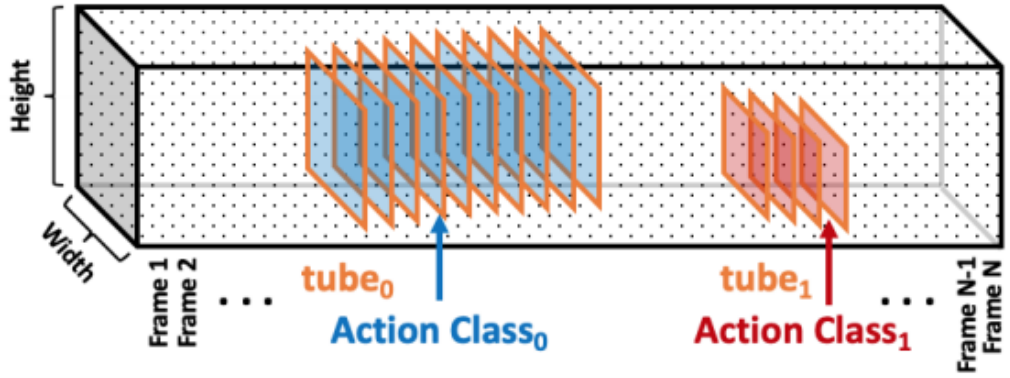
Temporal Action Localization/Detection (TAL/D)



Spatiotemporal Action Proposal (SAP)



Spatiotemporal Action Localization/Detection (SAL/D)



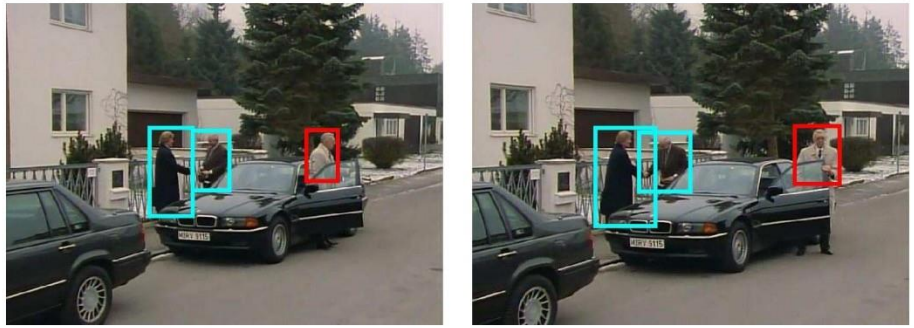
视频理解

- Temporal Action Localization
- 给定一个长视频序列，识别与不同动作对应的帧
- 可以使用类似 Faster R-CNN 的架构：先生成 temporal proposals，然后进行分类

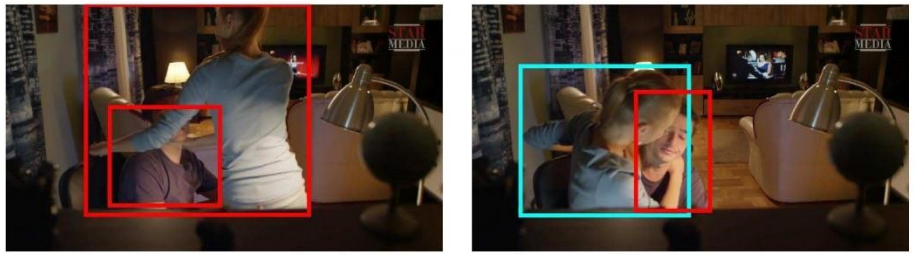
视频理解

- Spatio-Temporal Detection
- 给定一个长视频序列，检测空间和时间中的所有人，并对他们正在进行的活动进行分类

clink glass → drink



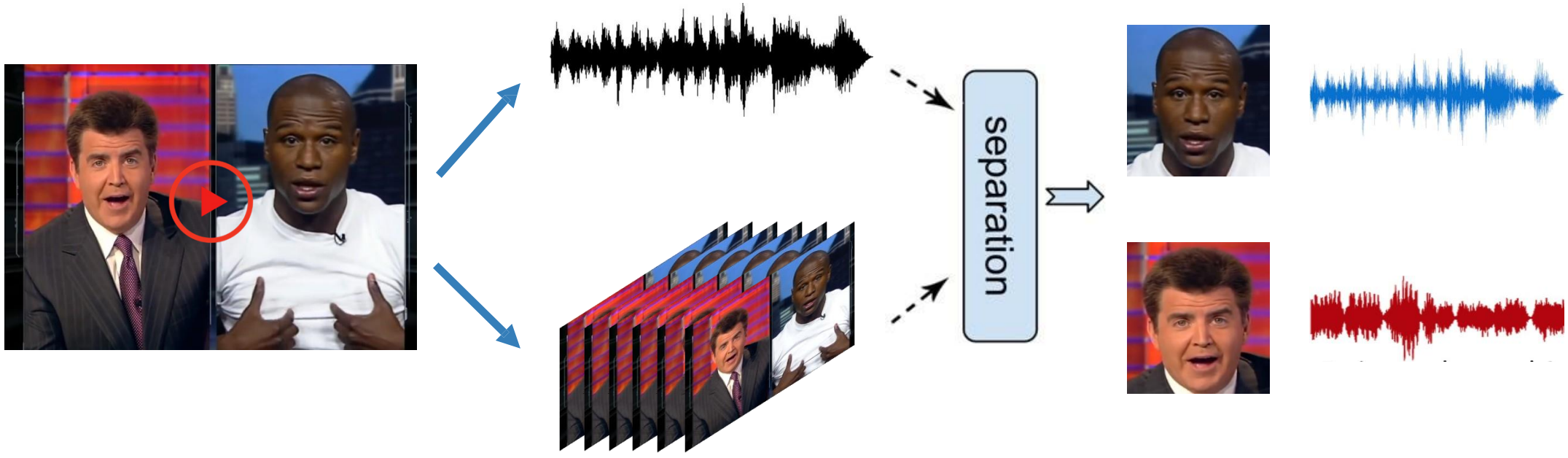
open → close



grab (a person) → hug

look at phone → answer phone

- Visually-guided audio source separation



视频理解

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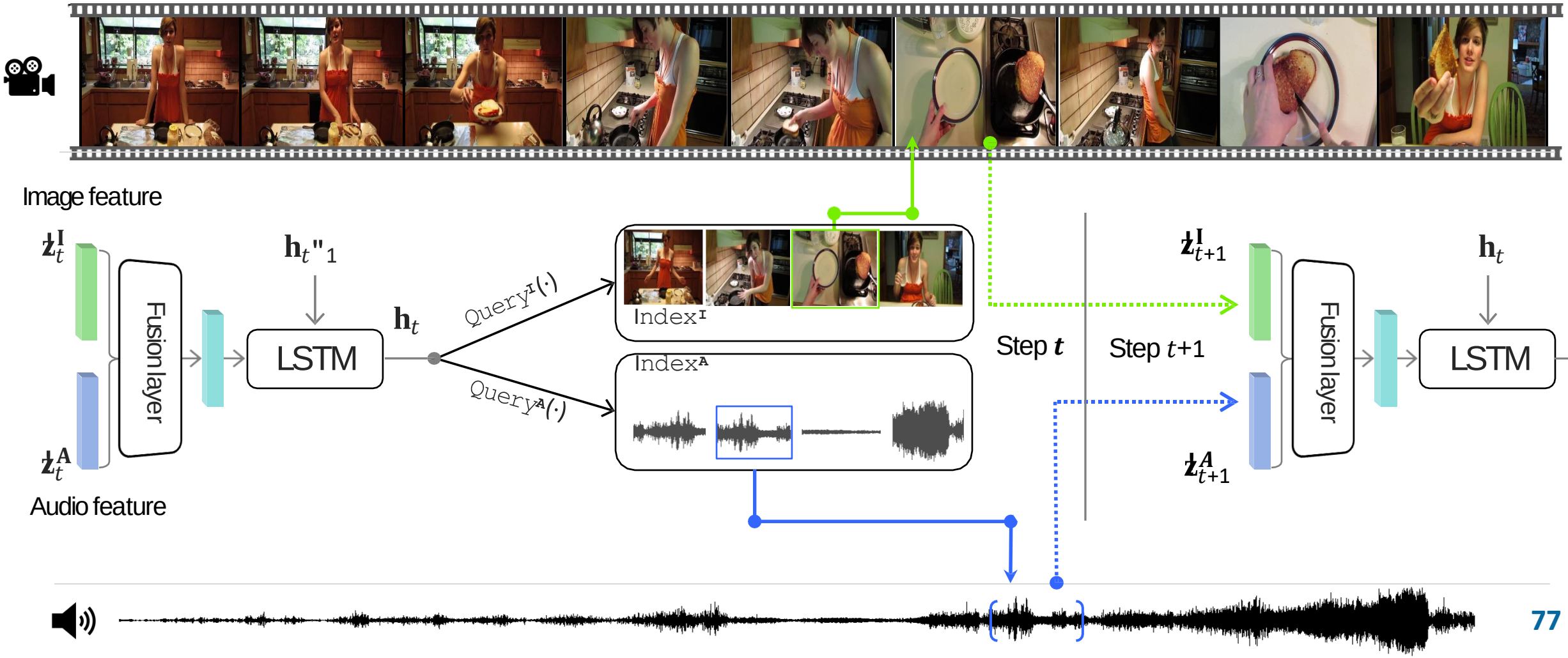
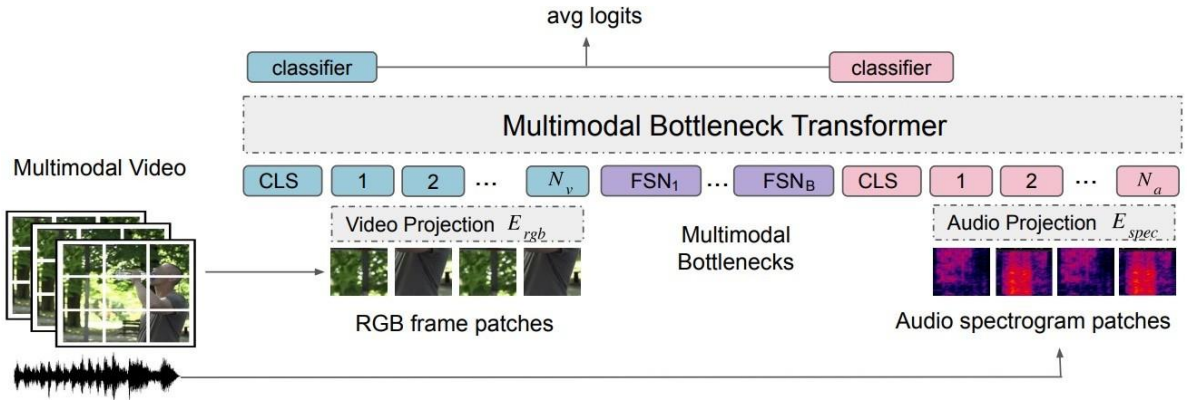
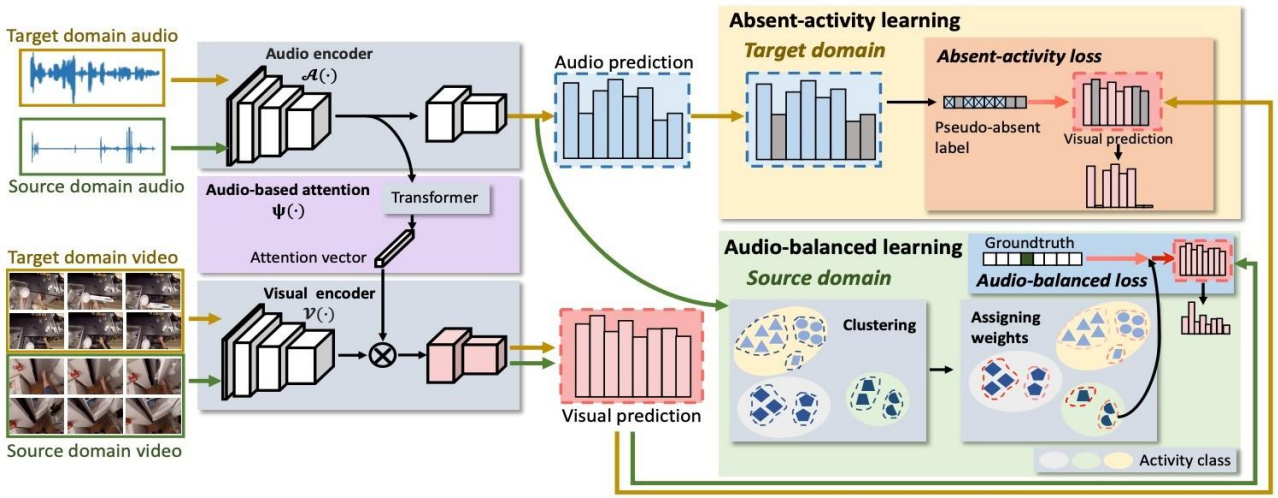
利用音频进行高效率的视频动作识别

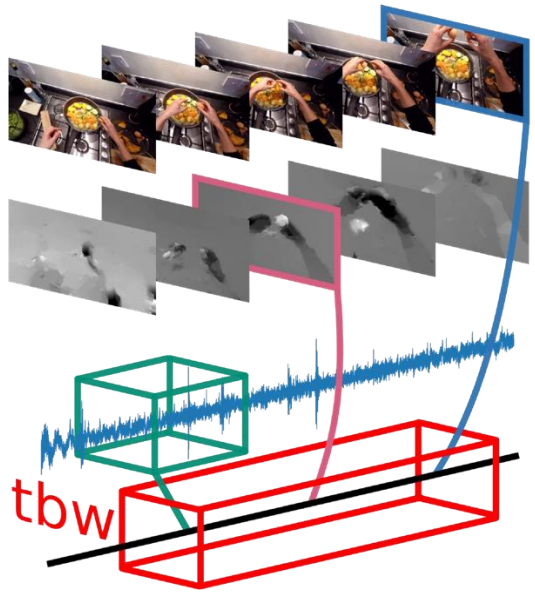


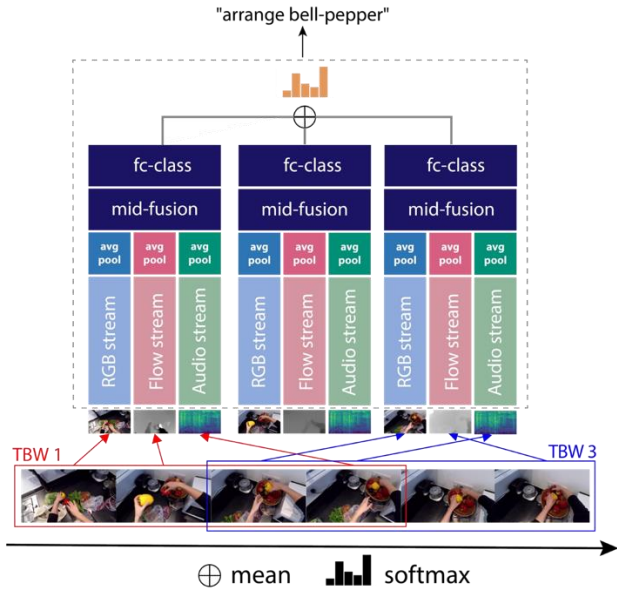
视频理解

多模态视频理解









视频理解

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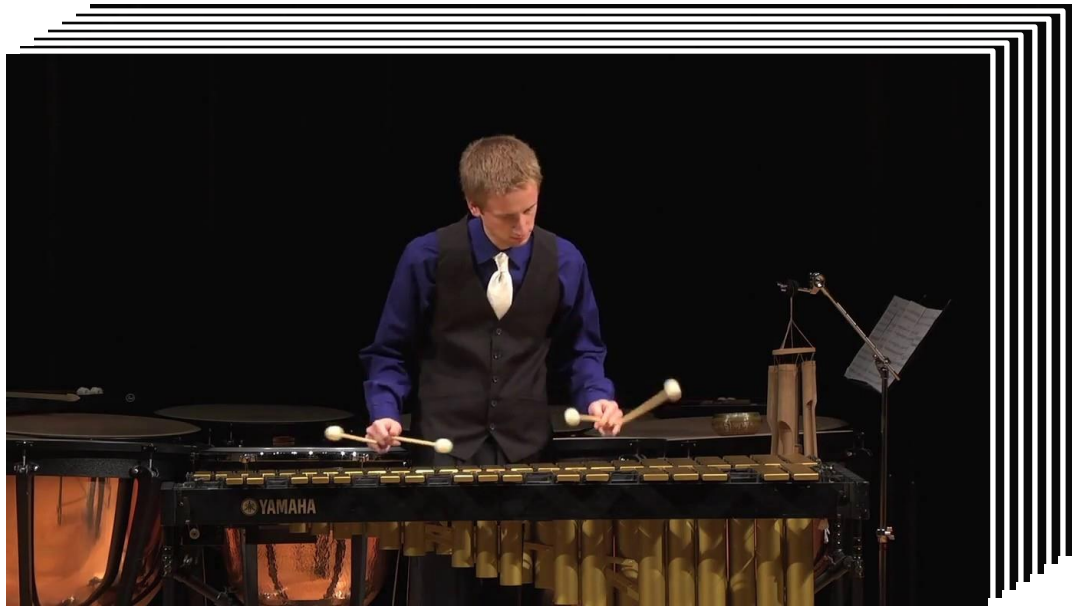
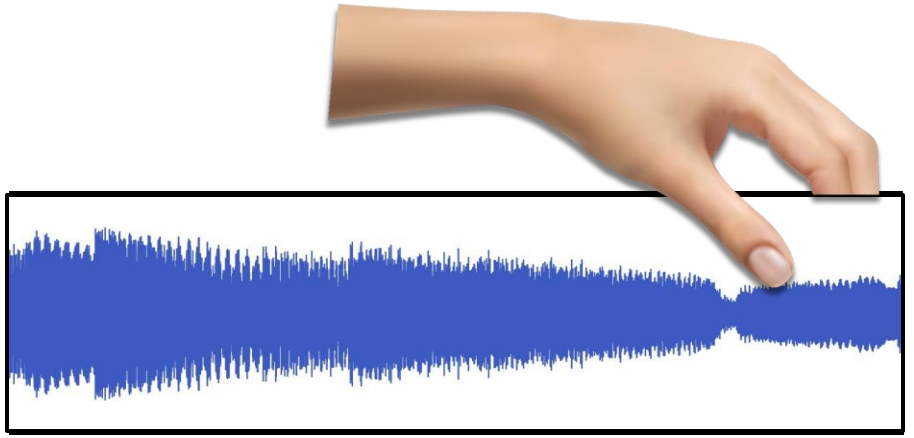
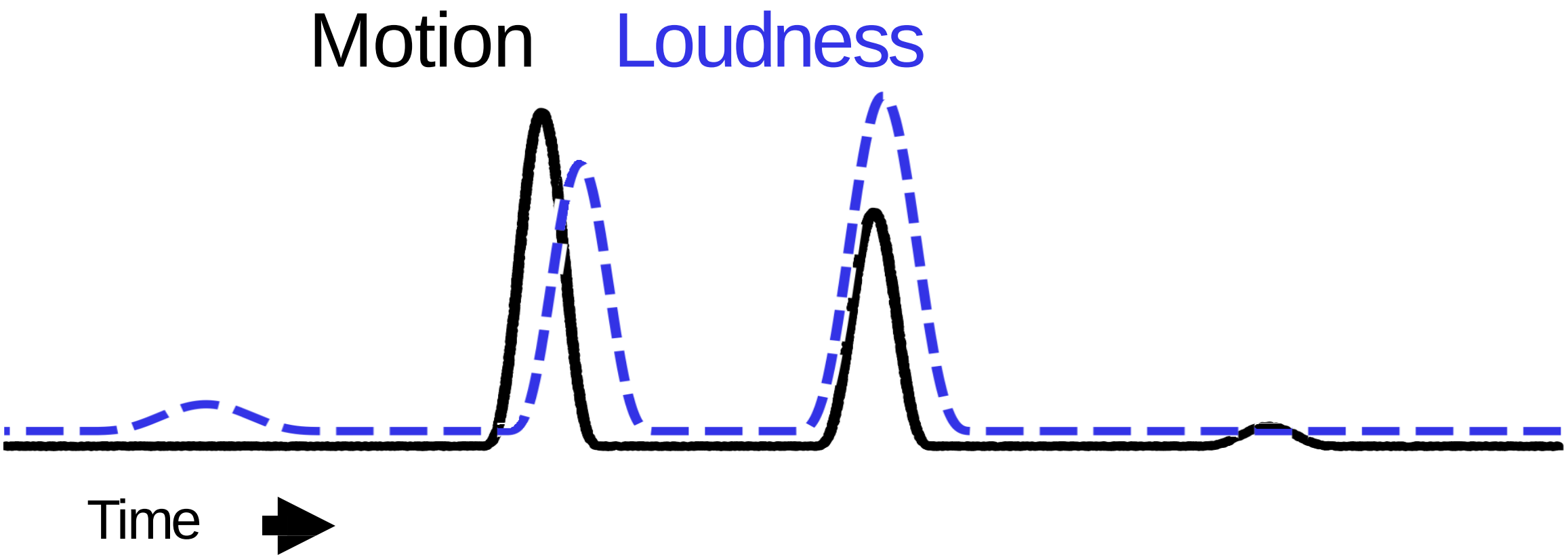
- 视听同步



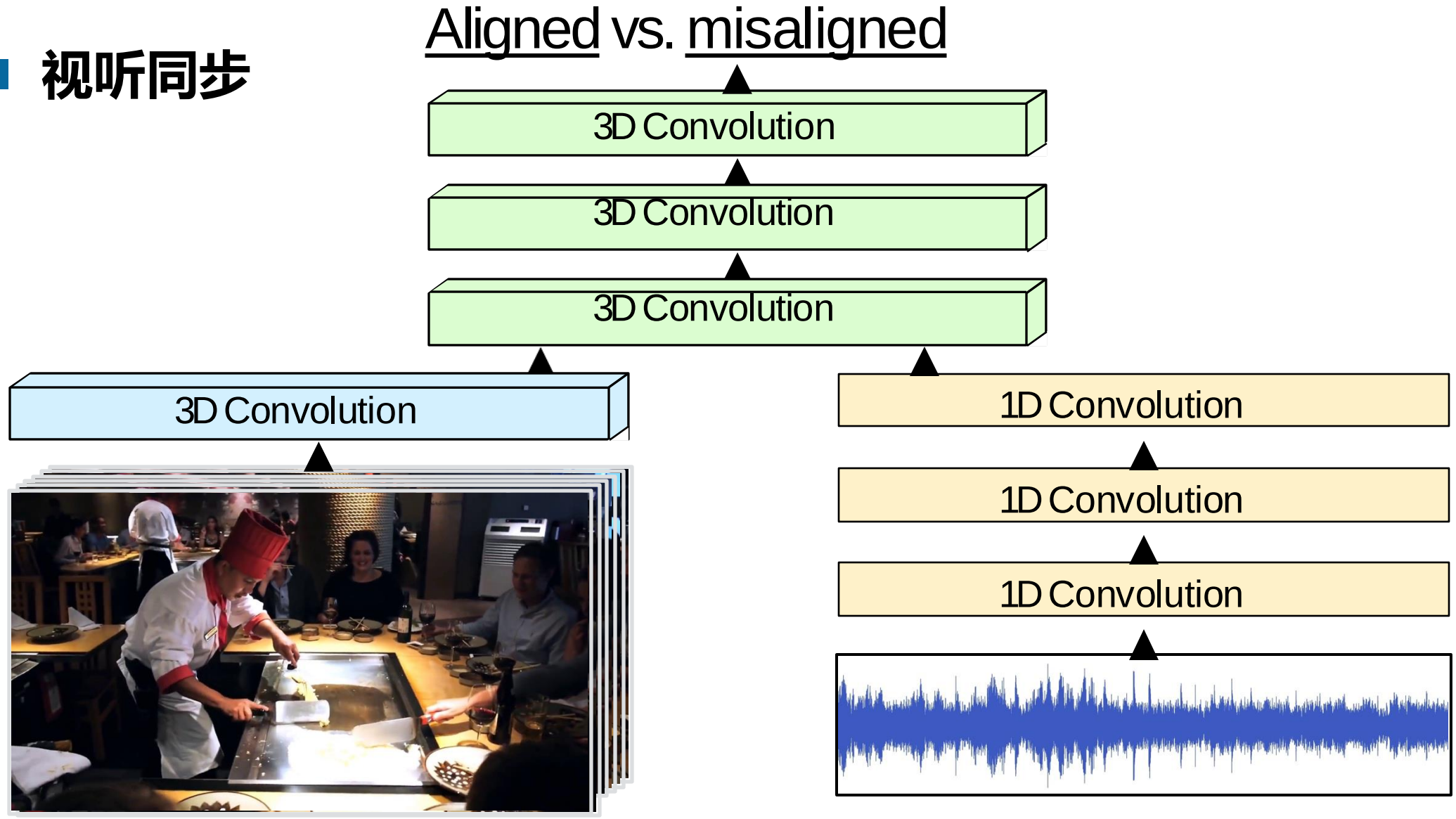


- 视听同步



视频理解

- 视听同步


Aligned vs. misaligned
3D Convolution
3D Convolution
3D Convolution
3D Convolution
1D Convolution
1D Convolution
1D Convolution